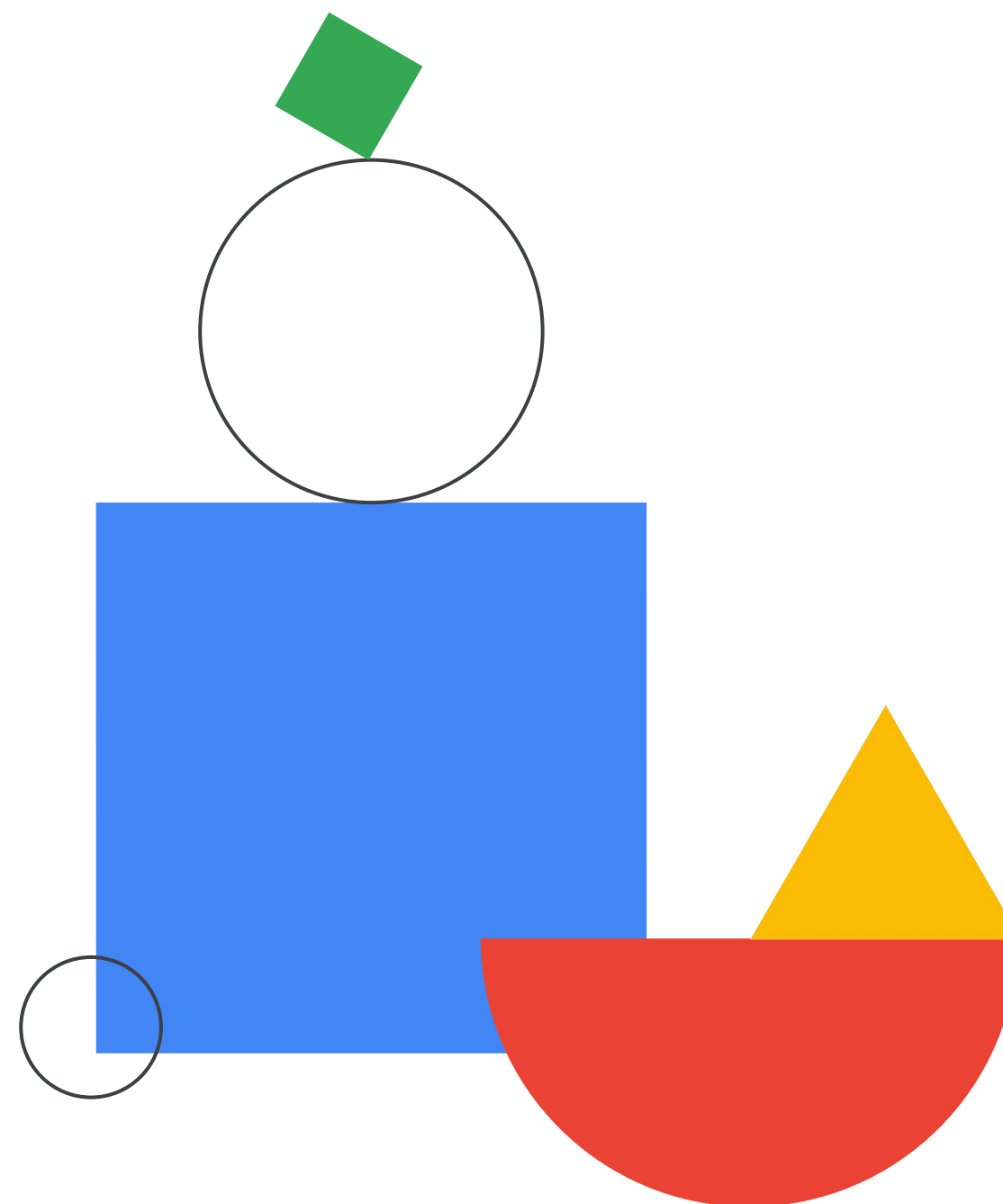
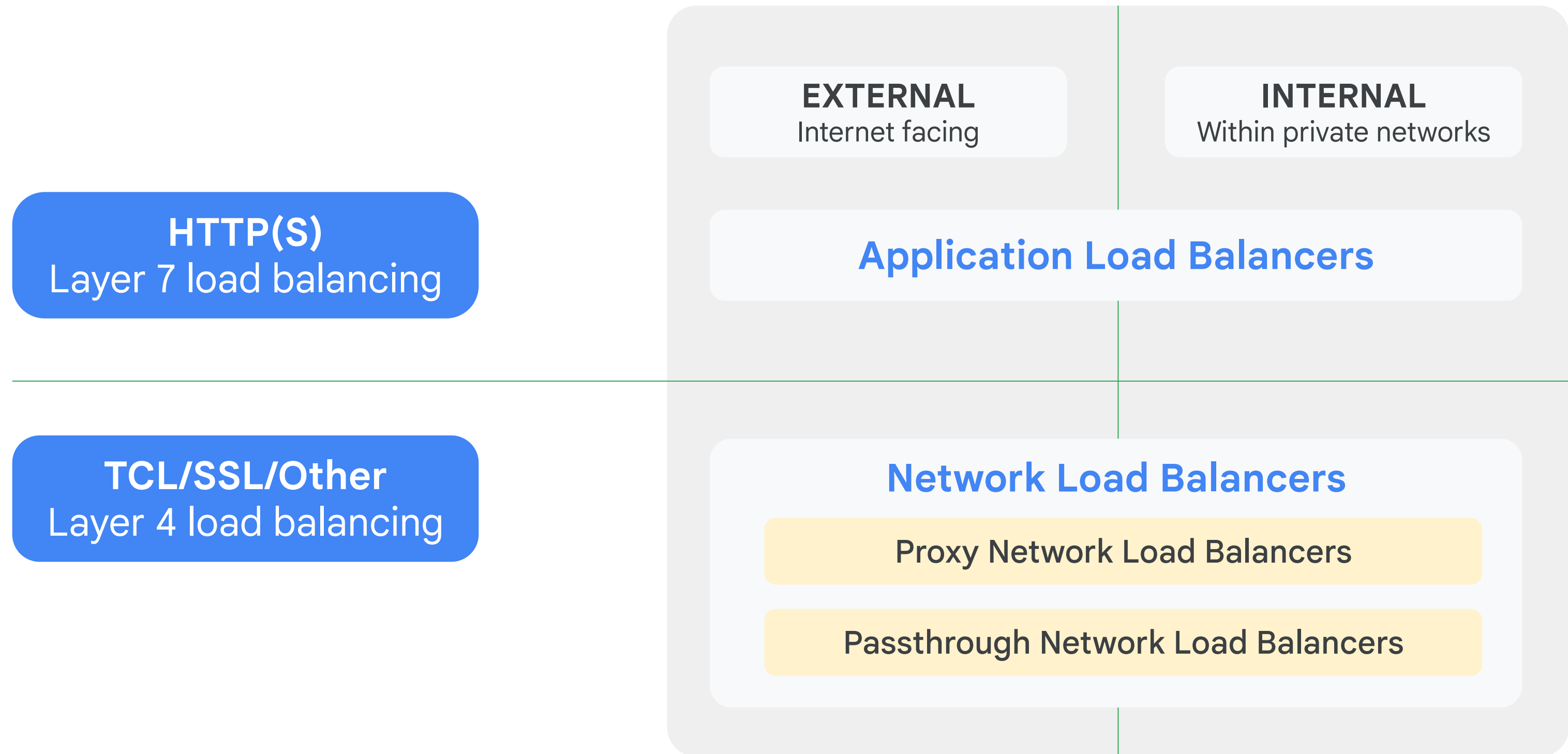




Load Balancing and Autoscaling

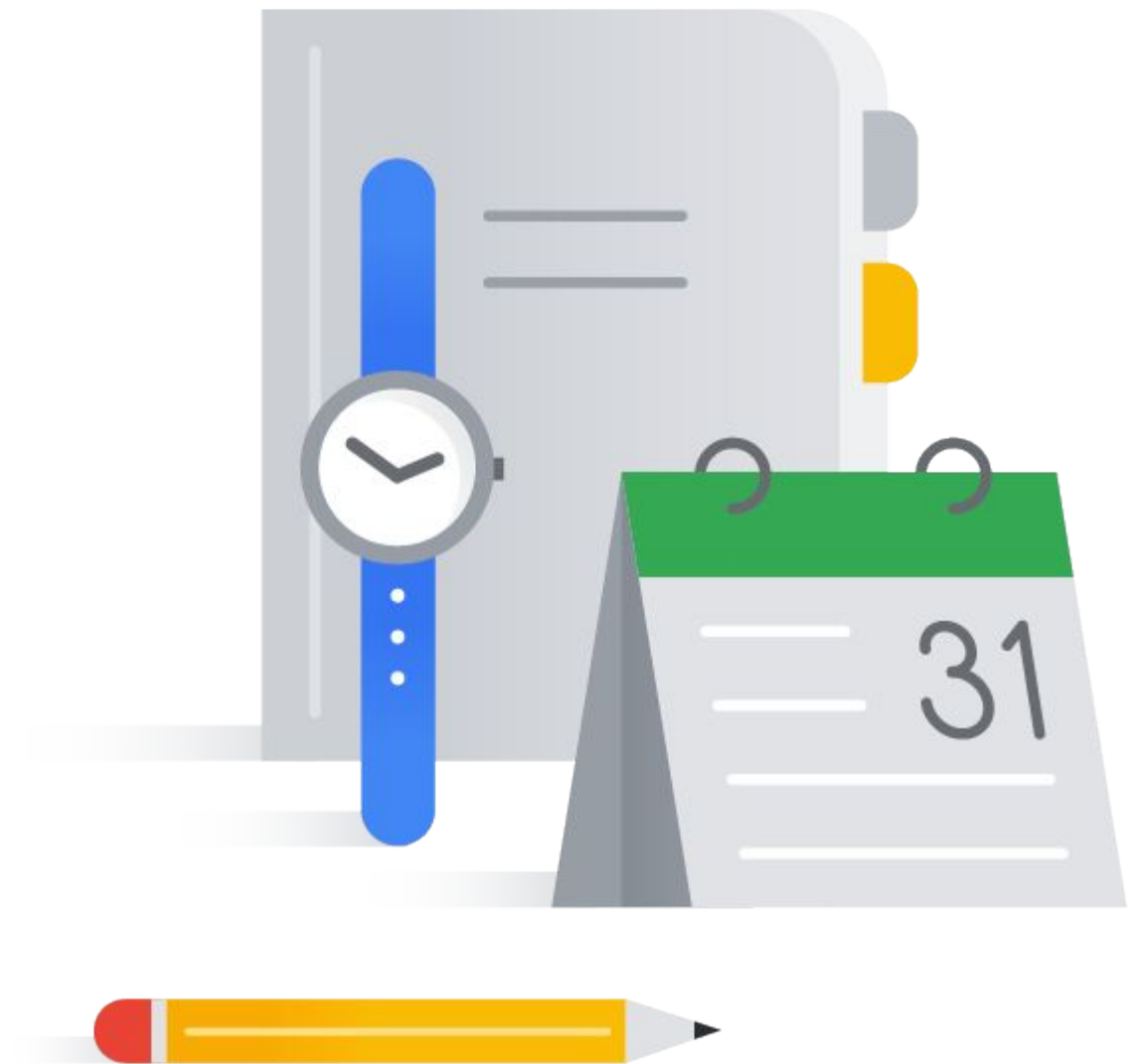


Types of load balancers



Agenda

- | | |
|----|---|
| 01 | Managed Instance Groups |
| 02 | Application Load Balancers
Lab: Configure an Application Load Balancer (HTTP) with Autoscaling |
| 03 | Cloud CDN |
| 04 | Network Load Balancing |
| 05 | Internal Load Balancing
Lab: Configure an Internal Network Load Balancer |
| 06 | Choosing a Load Balancer |

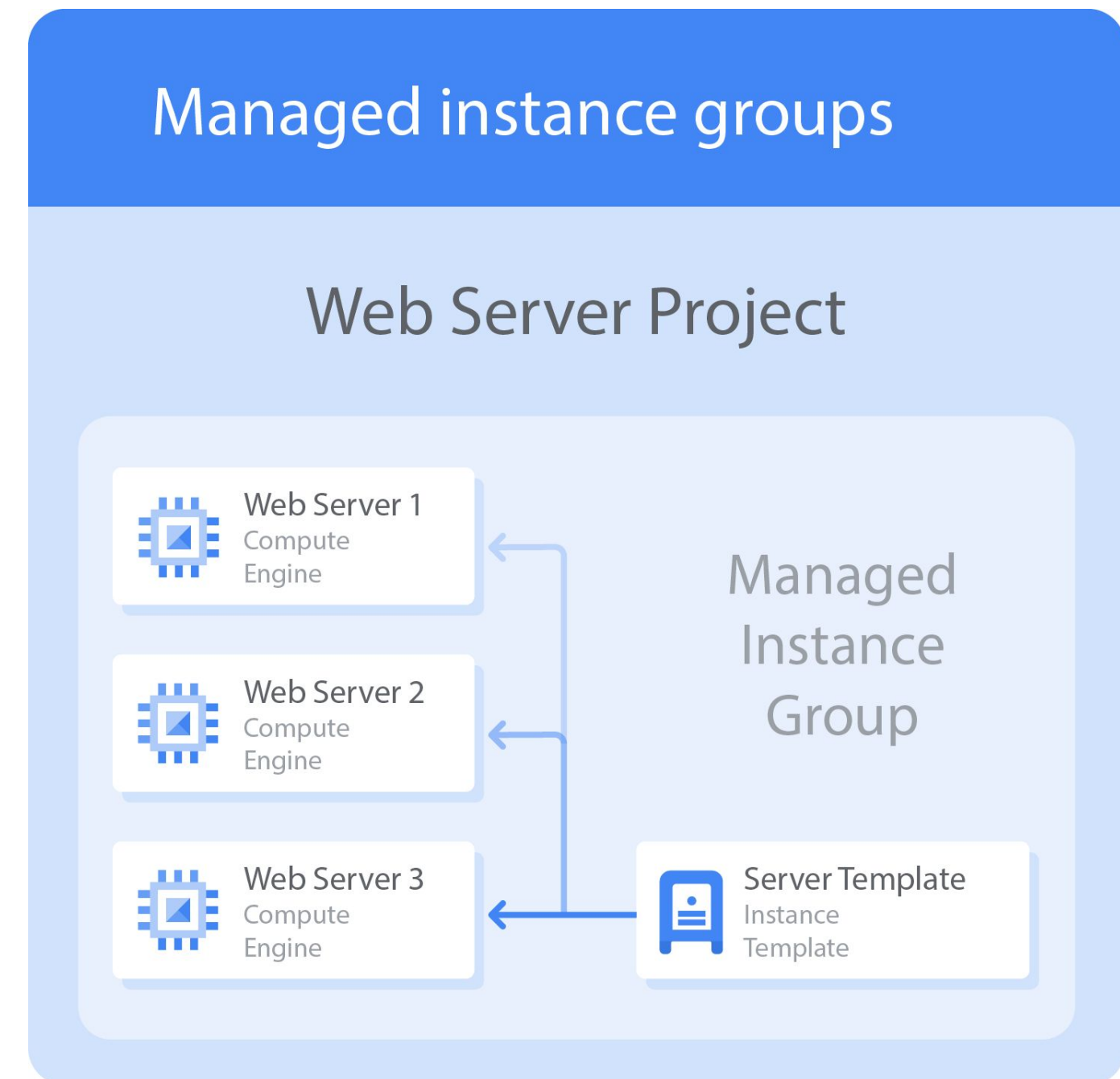




Managed Instance Groups

Managed instance groups

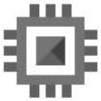
- Deploy identical instances based on instance template
- Instance group can be resized
- Manager ensures all instances are RUNNING
- Typically used with autoscaler
- Can be single zone or regional



Create an instance template

 CREATE INSTANCE TEMPLATE



 **Compute Engine**

Virtual machines

 VM instances

 Instance templates

Name *

mywebserver-template

MANAGE TAGS AND LABELS

Location

Global instance templates can be used in any region. To lower the impact of outages outside your region and gain data residency within your region, use a regional instance template. [Learn more](#)

☒ Global

☐ Regional (recommended)

Machine configuration

NEW: General-purpose machine series in Preview

Try the new N4 series, ideal for workloads that prioritize flexibility and cost-optimization

SIGN UP

General purpose

Compute optimized

Memory optimized

Storage optimized

NEW

GPUs

Machine types for common workloads, optimized for cost and flexibility

	Series	Description	vCPUs	Memory	Platform
☐	C3	Consistently high performance	4 - 176	8 - 1,408 GB	Intel Sapphire Rapids
☐	C3D	Consistently high performance	4 - 360	8 - 2,880 GB	AMD Genoa
☒	E2	Low cost, day-to-day computing	0.25 - 32	1 - 128 GB	Based on availability
☐	N2	Balanced price & performance	2 - 128	2 - 864 GB	Intel Cascade and Ice Lake
☐	N2D	Balanced price & performance	2 - 224	2 - 896 GB	AMD EPYC
☐	T2A	Scale-out workloads	1 - 48	4 - 192 GB	Ampere Altra Arm
☐	T2D	Scale-out workloads	1 - 60	4 - 240 GB	AMD EPYC Milan
☐	N1	Balanced price & performance	0.25 - 96	0.6 - 624 GB	Intel Skylake

Machine type

Choose a machine type with preset amounts of vCPUs and memory that suit most workloads. Or, you can create a custom machine for your workload's particular needs. [Learn more](#)

PRESET

CUSTOM

e2-medium (2 vCPU, 1 core, 4 GB memory)

 vCPU

1-2 vCPU (1 shared core)

 Memory

4 GB

ADVANCED CONFIGURATIONS

Boot disk

Name

mywebserver-template

Type

New balanced persistent disk

Size

10 GB

License type

Free

Image

 Debian GNU/Linux 11 (bullseye)

CHANGE

Identity and API access

Service accounts

Service account

Compute Engine default service account

Requires the Service Account User role (roles/iam.serviceAccountUser) to be set for users who want to access VMs with this service account. [Learn more](#)

Access scopes

☒ Allow default access

☐ Allow full access to all Cloud APIs

☐ Set access for each API

Firewall

Add tags and firewall rules to allow specific network traffic from the Internet

☐ Allow HTTP traffic

☐ Allow HTTPS traffic

☐ Allow Load Balancer Health Checks

Advanced options

Networking, disks, security, management, sole-tenancy

Google Cloud

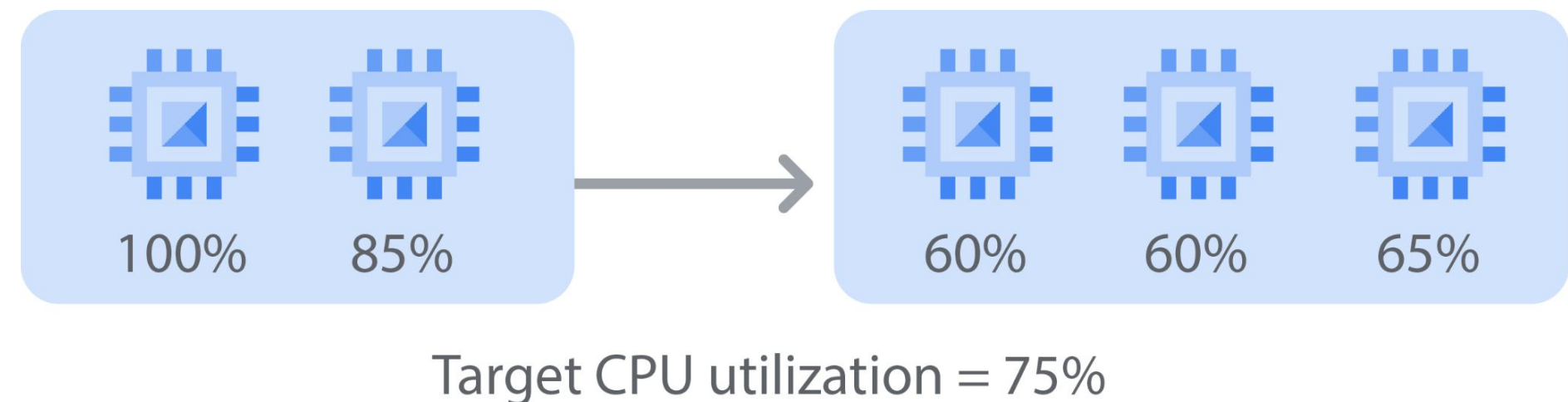
Managed instance groups offer autoscaling capabilities

Dynamically add/remove instances:

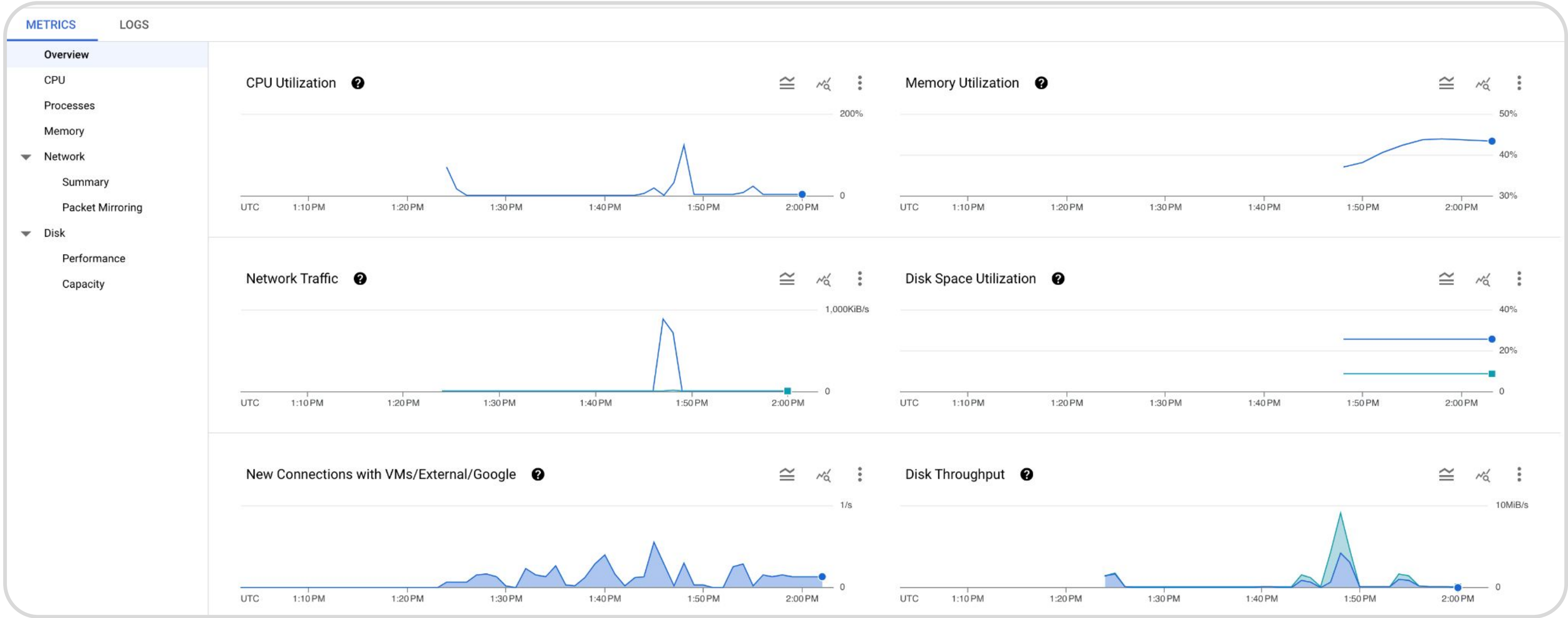
- Increases in load
- Decreases in load

Autoscaling policy:

- CPU utilization
- Load balancing capacity
- Monitoring metrics
- Queue-based workload
- Schedule-based



VM graph helps set CPU utilization



Create a health check

Health checking mechanisms determine whether VM instances respond properly to traffic. You cannot create a legacy health check using this page. For more information, refer to the [Health Checks Concepts](#) documentation.

Name *

Lowercase, no spaces.

Description

Scope

☒ Global

☐ Regional

Protocol

TCP

Port *

80

Proxy protocol

NONE

Request

Response

Logs

☐ On

☒ Off

Turning on Health check logs can increase costs in Logging.

Health criteria

Define how health is determined: how often to check, how long to wait for a response, and how many successful or failed attempts are decisive

Check interval *

5

seconds

Timeout *

5

seconds

Healthy threshold *

2

consecutive successes

Unhealthy threshold *

2

consecutive failures

Elapsed time (seconds)	Event duration (seconds)	
1	1	wait
2	2	wait
3	3	wait
4	4	wait
5	5	wait
6	1	health check #1 starts
7	2	wait
8	3	wait
9		health check #1 fails
10		wait
11	1	health check #2 starts
12	2	wait
13	3	wait
14		health check #2 fails
15		Unhealthy threshold reached

Configuring stateful IP addresses

Preserve the unique state of each MIG VM instance on machine restart, recreation, auto-healing, or update event. Useful in the following scenarios:

- ✓ IP address to remain static after it has been assigned.
- ✓ Configuration depends on specific IP addresses.
- ✓ Server is accessed through a dedicated static IP address.
- ✓ Migrate workloads without changing network configuration.

- > Configure IP addresses as stateful for all existing and future instances in the group.
- > Update the existing stateful configuration for IP addresses.

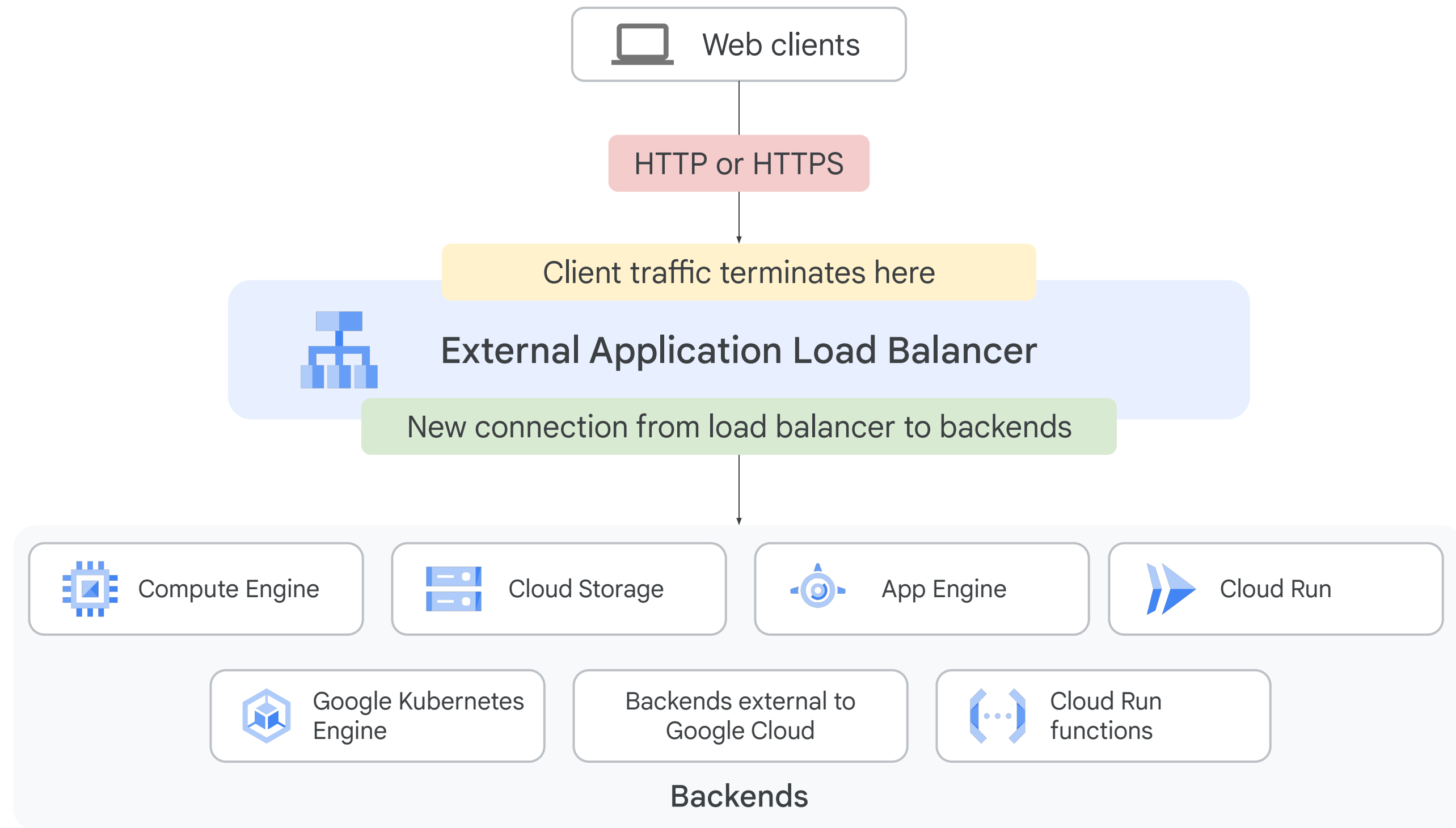


Application Load Balancing

Overview of an Application Load Balancer

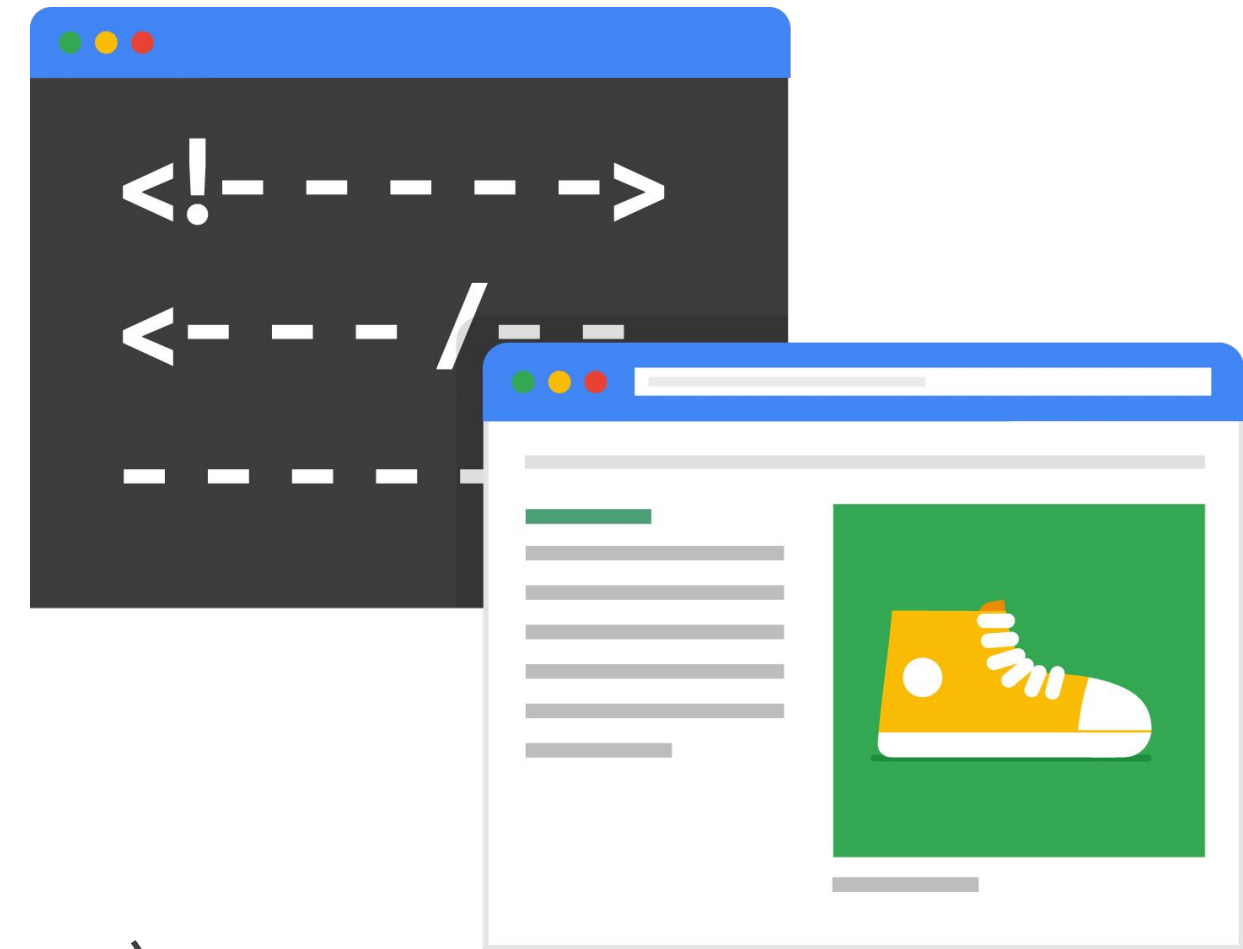
Deployment mode	Network service tier	Load balancing scheme	IP address	Frontend ports
Global external	Premium Tier	EXTERNAL_MANAGED	IPv4 IPv6	Can reference exactly one port from 1-65535
Regional external	Premium or Standard Tier	EXTERNAL_MANAGED	IPv4	
Classic	Global in Premium Tier Regional in Standard Tier	EXTERNAL	IPv4 IPv6 (requires Premium Tier)	

Architecture of an external Application Load Balancer

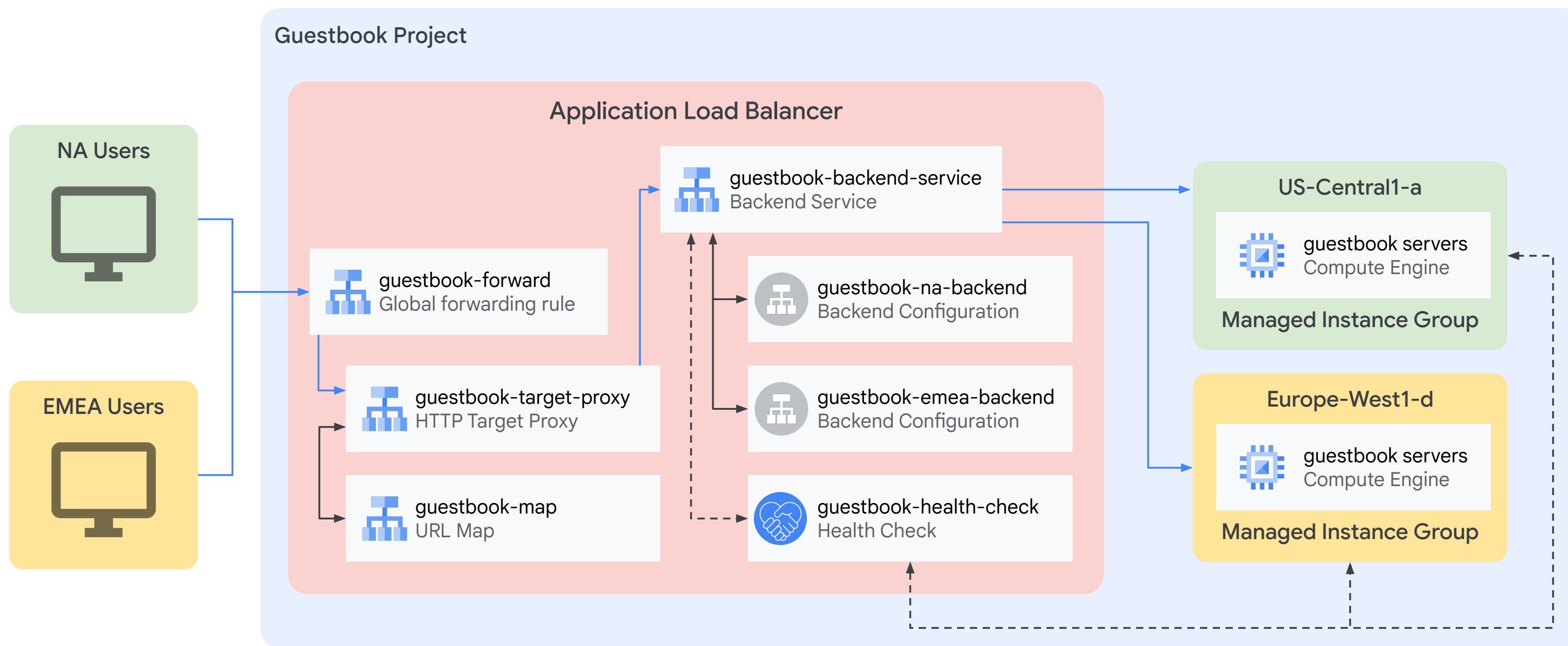


Backend services

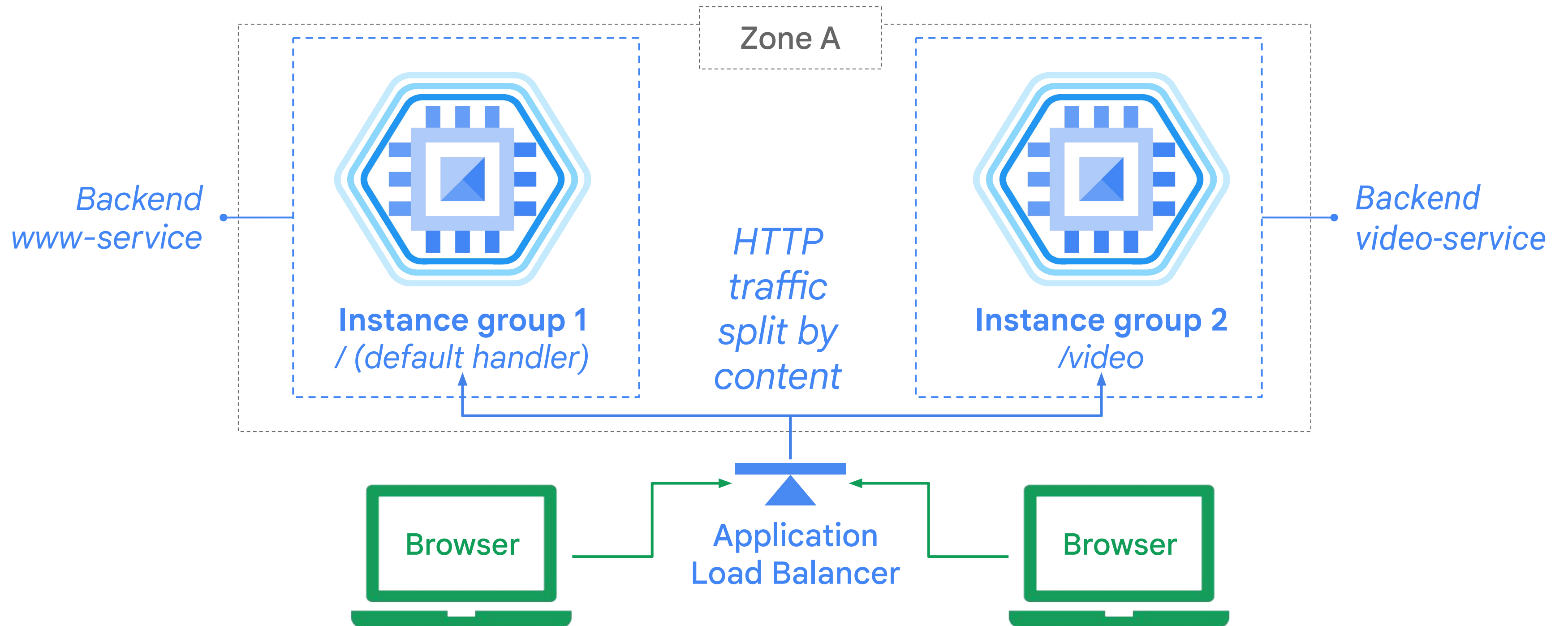
- Health check
- Session affinity (optional)
- Time out setting (30-sec default)
- One or more backends
 - An instance group (managed or unmanaged)
 - A balancing mode (CPU utilization or RPS)
 - A capacity scaler (ceiling percentage of CPU/Rate targets)



Application Load Balancing resources

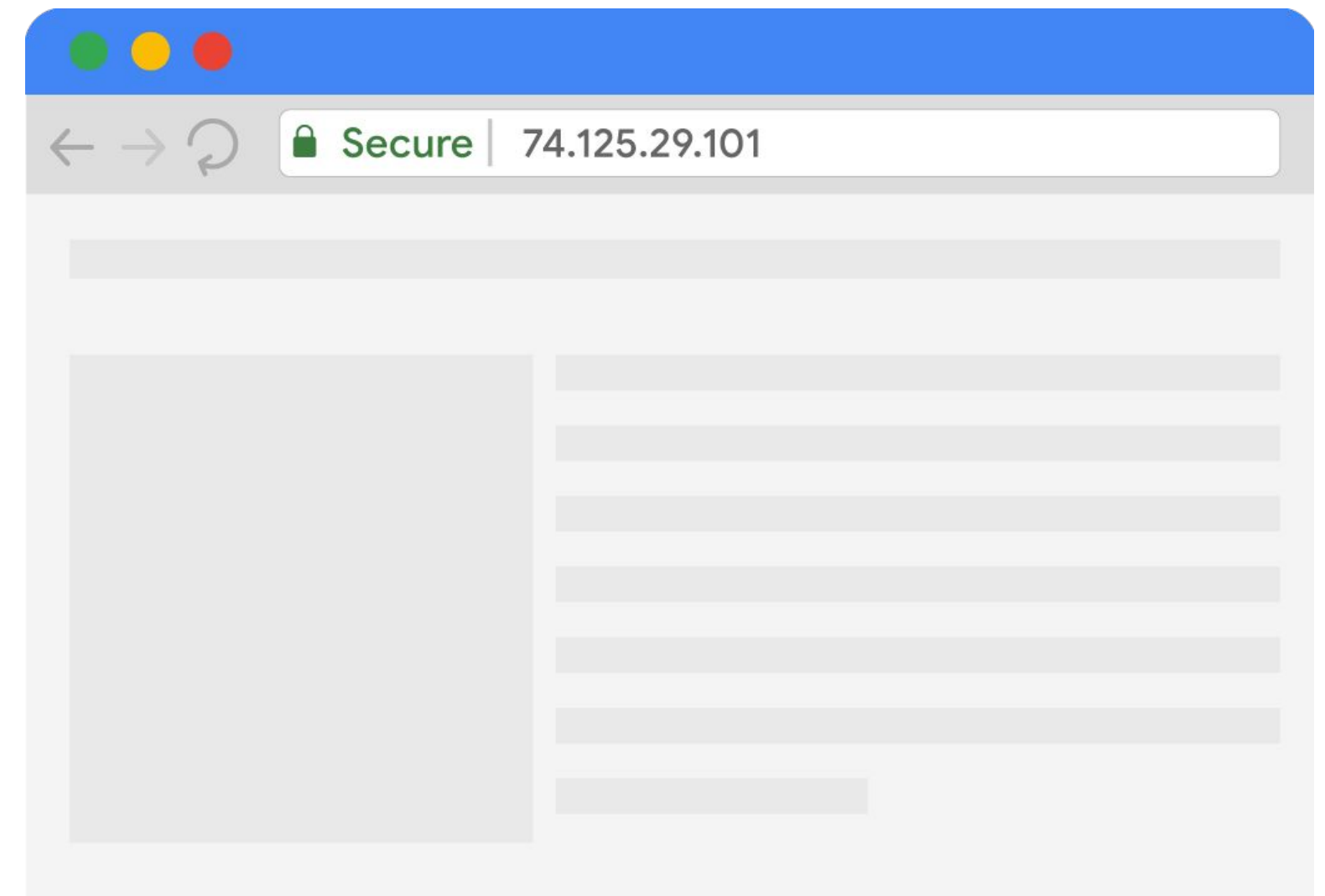


Example: Content-based load balancing



Application Load Balancing - Target HTTPS proxy

- Target HTTP(S) proxy
- One signed SSL certificate installed (minimum)
- Client SSL session terminates at the load balancer
- Support the QUIC transport layer protocol

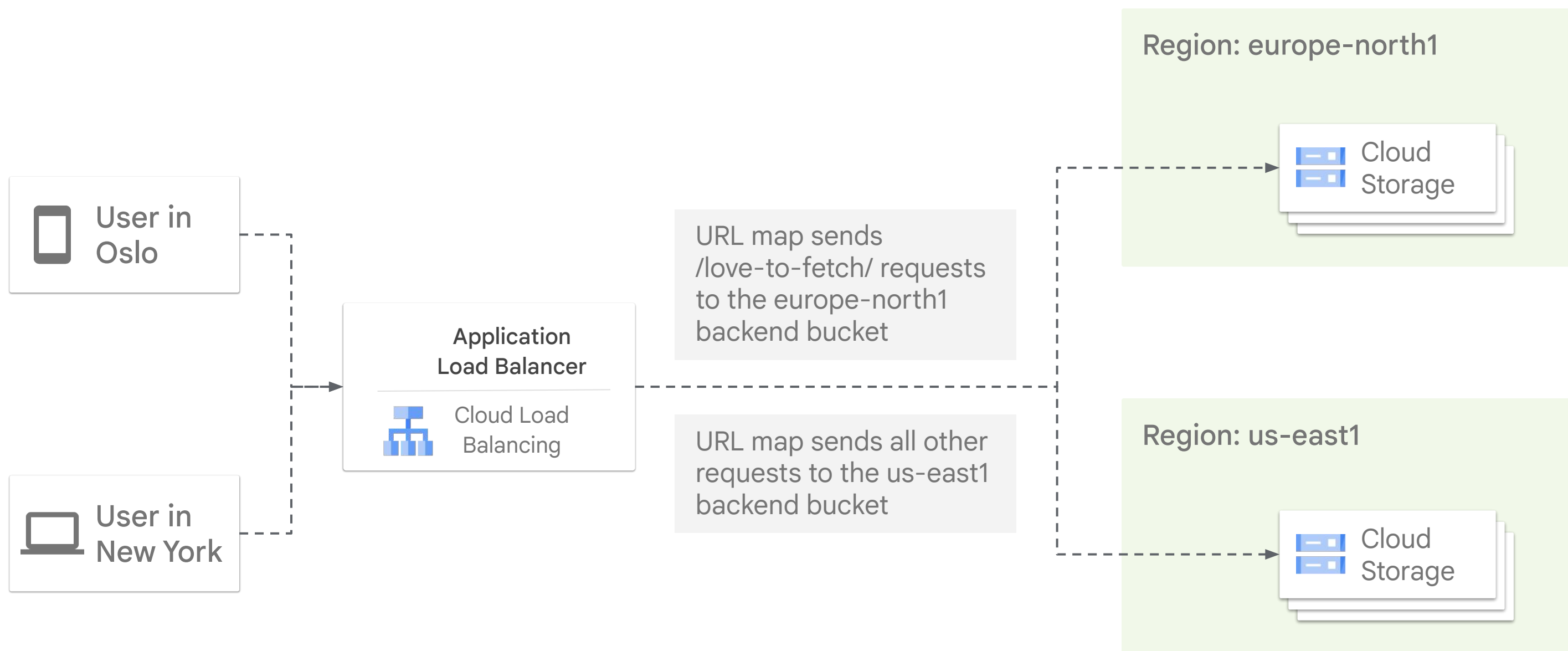


SSL certificates

- Required for Application Load Balancing
- Up to 15 SSL certificates (per target proxy)
- Create an SSL certificate resource



Backend buckets



Network endpoint groups (NEG)

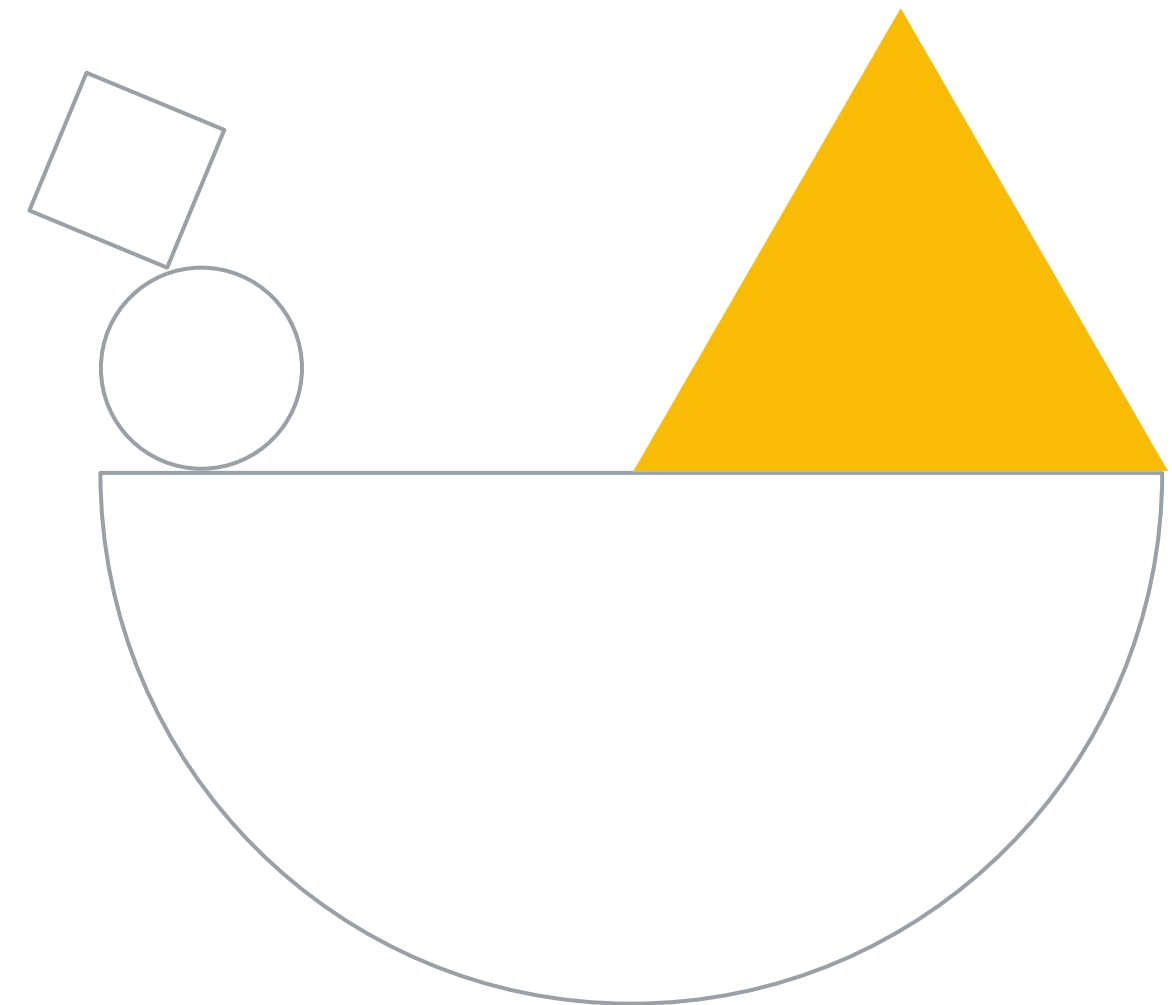
A network endpoint group (NEG) is a configuration object that specifies a group of backend endpoints or services.

There are four types of NEG:

- Zonal
- Internet
- Serverless
- Hybrid connectivity

Lab Intro

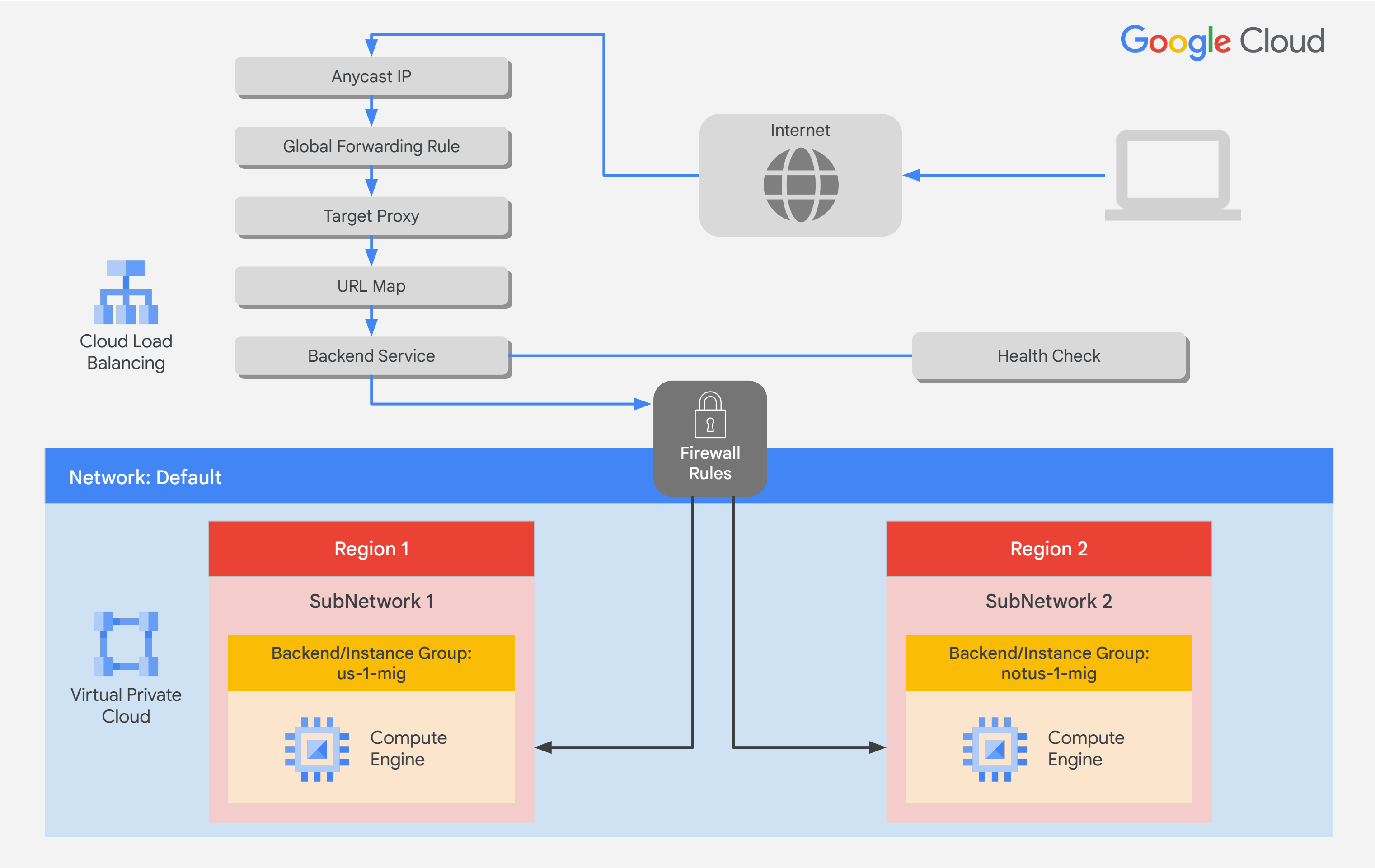
Configure an Application Load
Balancer (HTTP) with Autoscaling



Lab objectives

- 01 Create HTTP and health check firewall rules
- 02 Create a custom image for a web server
- 03 Create an instance template based on the custom image
- 04 Create two managed instance groups
- 05 Configure an Application Load Balancer with IPv4 and IPv6
- 06 Stress test an Application Load Balancer







Cloud CDN

Regions and CDN nodes

Oregon
Salt Lake City
Los Angeles
Phoenix
Queretaro
Iowa
Toronto
Montréal
N. Virginia
Columbus
Dallas
S. Carolina

São Paulo
Santiago

Norway
Finland
Sweden
Netherlands
Berlin
London
Belgium
Paris
Turin
Madrid
Frankfurt
Zurich
Milan
Warsaw

Tel Aviv
Dammam
Doha
Mumbai
Delhi
Johannesburg

Seoul
Tokyo
Osaka
Hong Kong
Taiwan
Singapore
Jakarta
Melbourne
Sydney

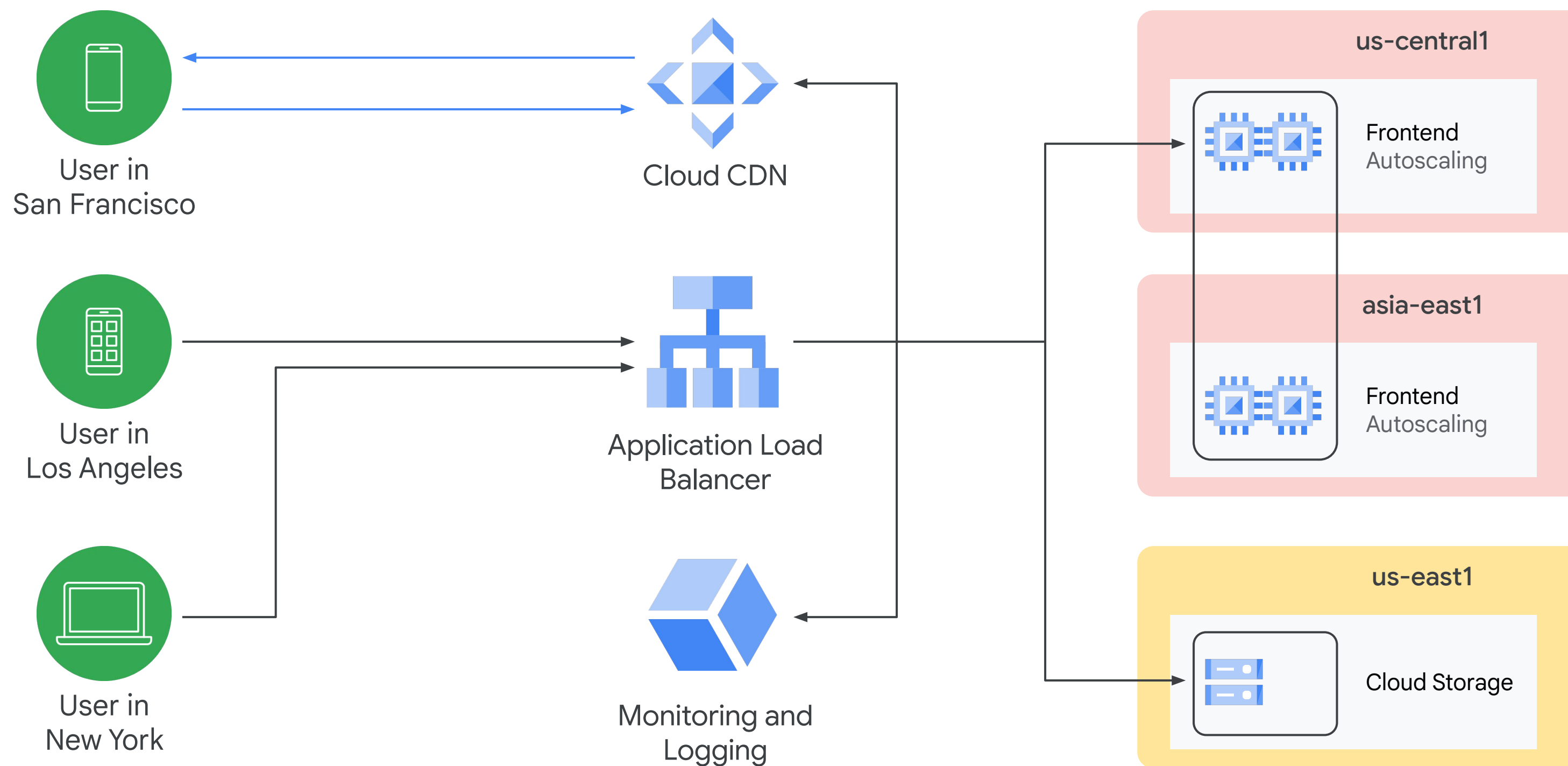
*Exception: region has 4 zones.

● CDN

● Current region with 3 zones

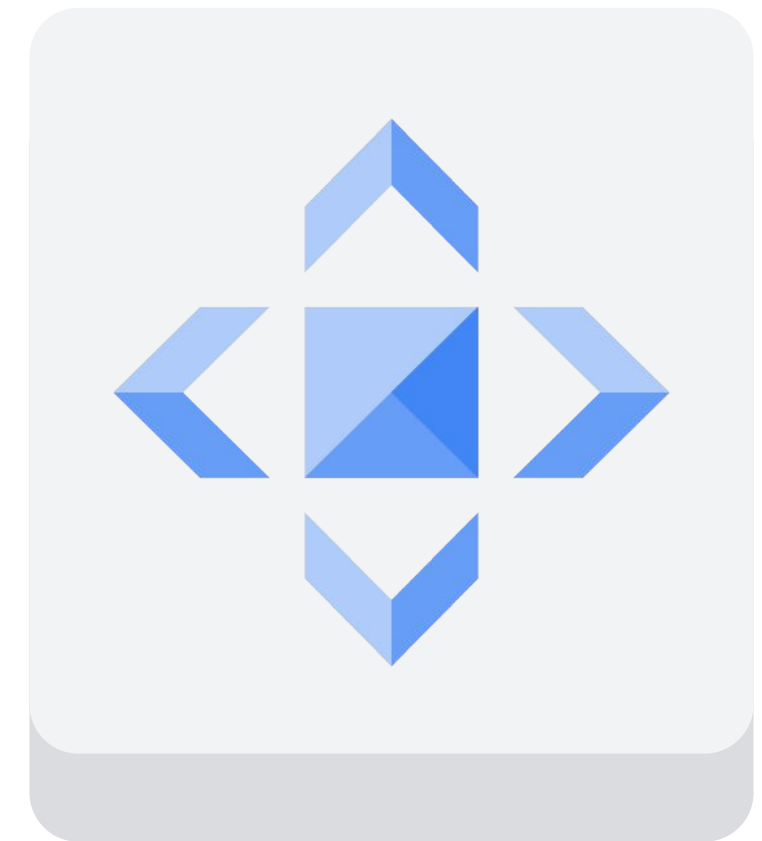
○ Future region with 3 zones

Caching content with Cloud CDN



Cloud CDN cache modes

- Cache modes control the factors that determine whether or not Cloud CDN caches your content.
- Cloud CDN offers three cache modes:
 - USE_ORIGIN_HEADERS
 - CACHE_ALL_STATIC
 - FORCE_CACHE_ALL

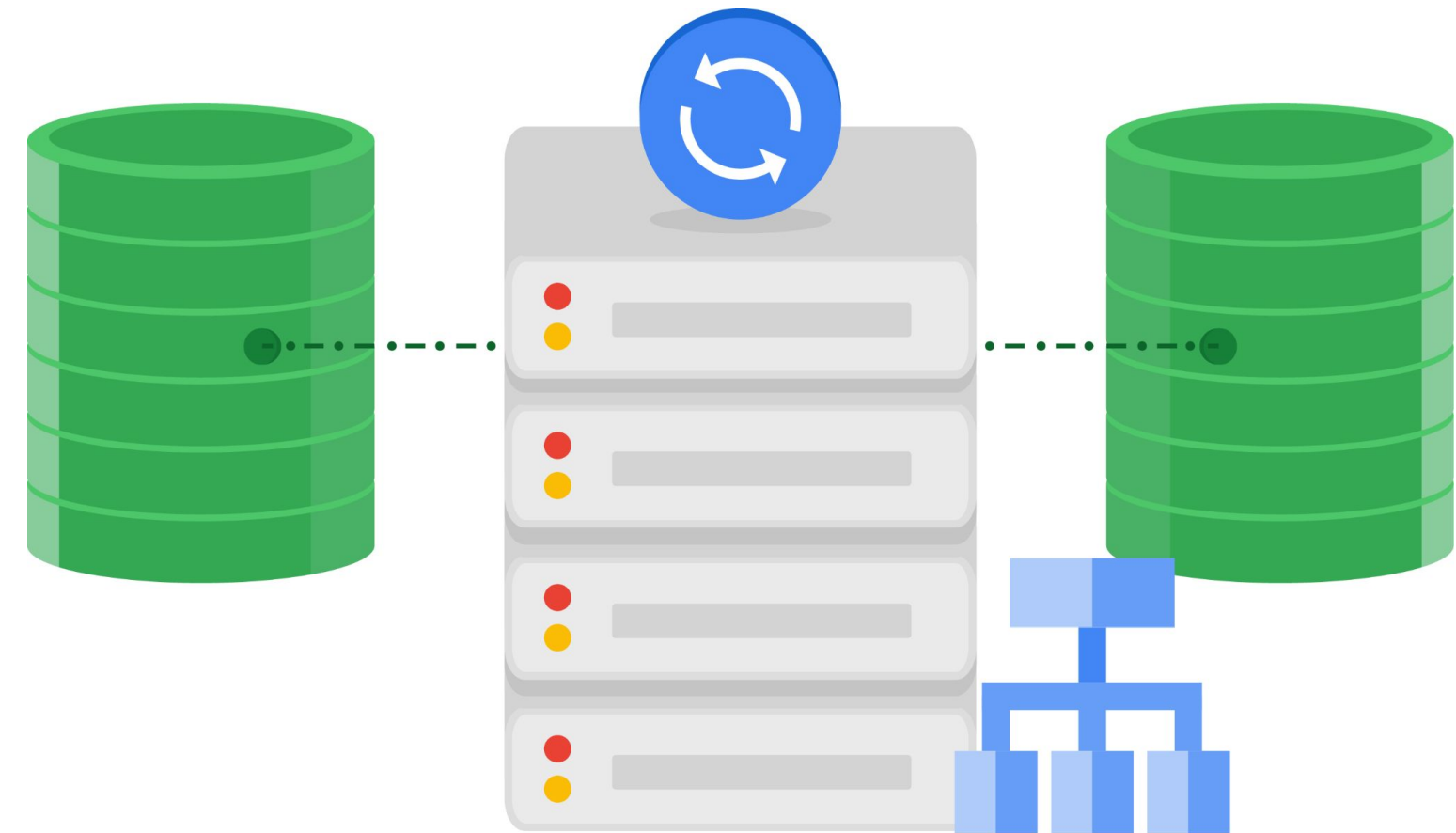




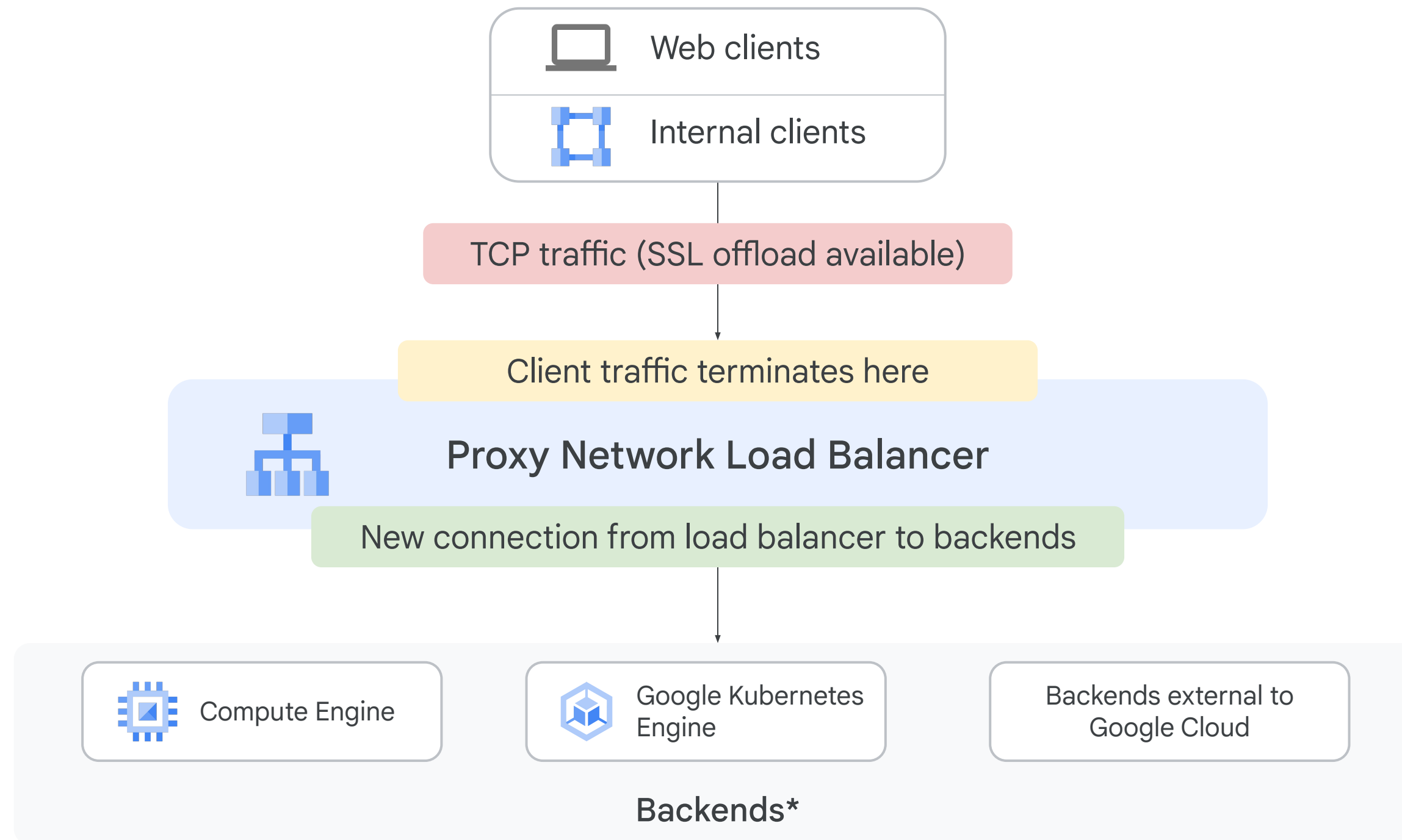
Network Load Balancing

Network load balancing

- Architecture:
 - Proxy
 - Passthrough
- Traffic:
 - TCP/SSL ports
 - UDP, ESP, GRE
 - ICMP, ICMPv6

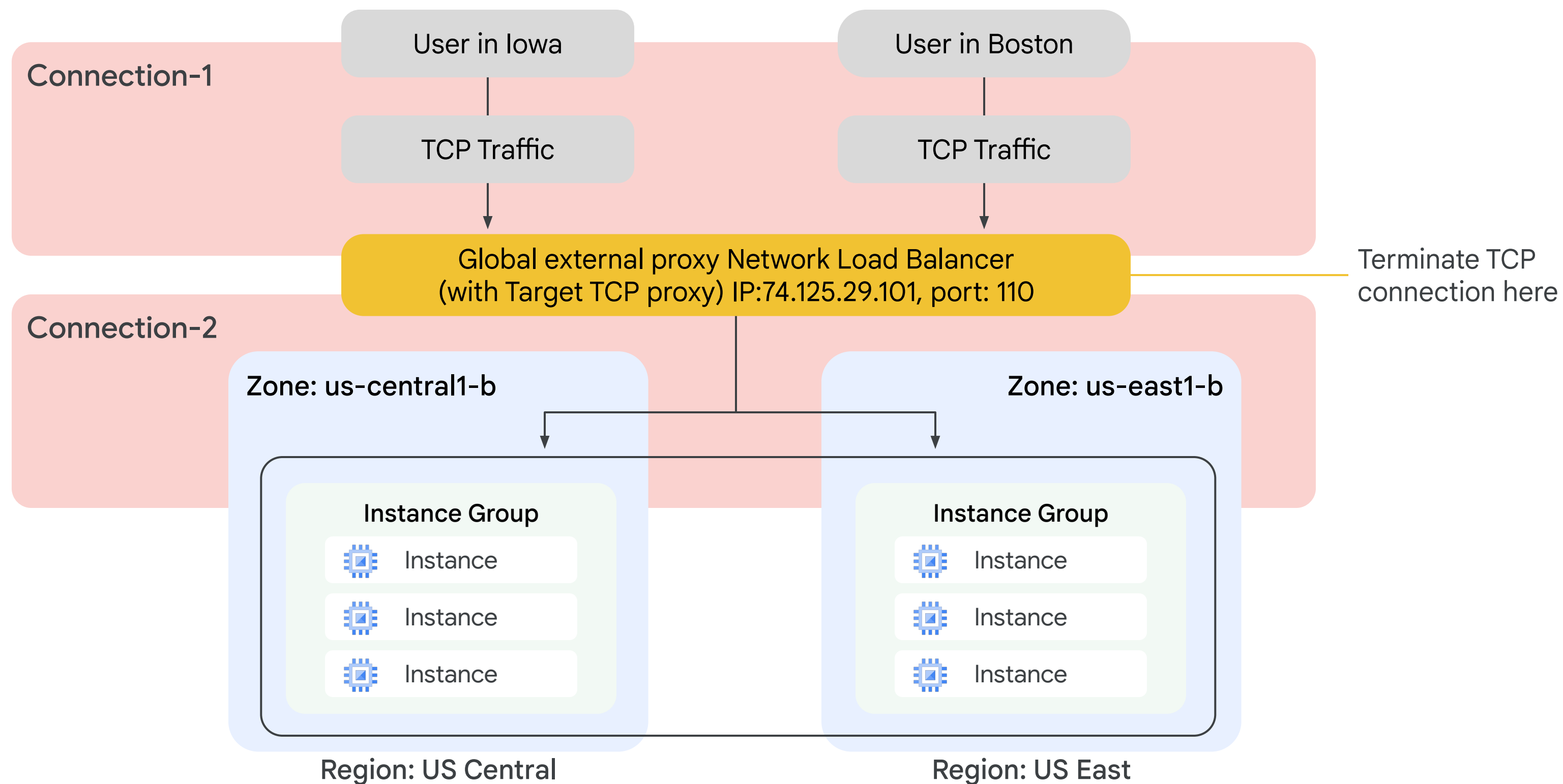


Architecture of a Proxy Network Load Balancer

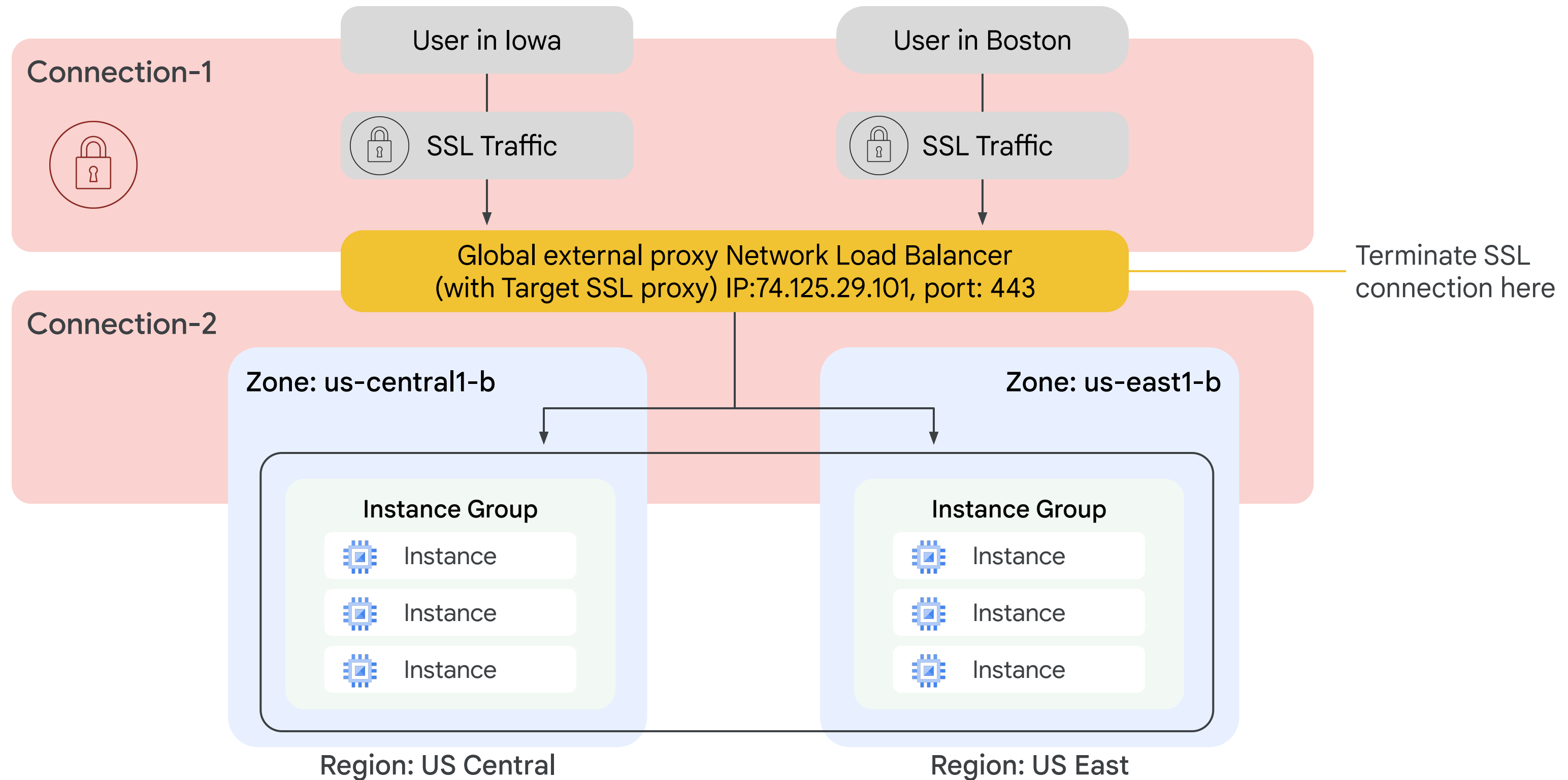


*Backend support differs depending on the deployment mode of the load balancer (internal or external, global or regional).

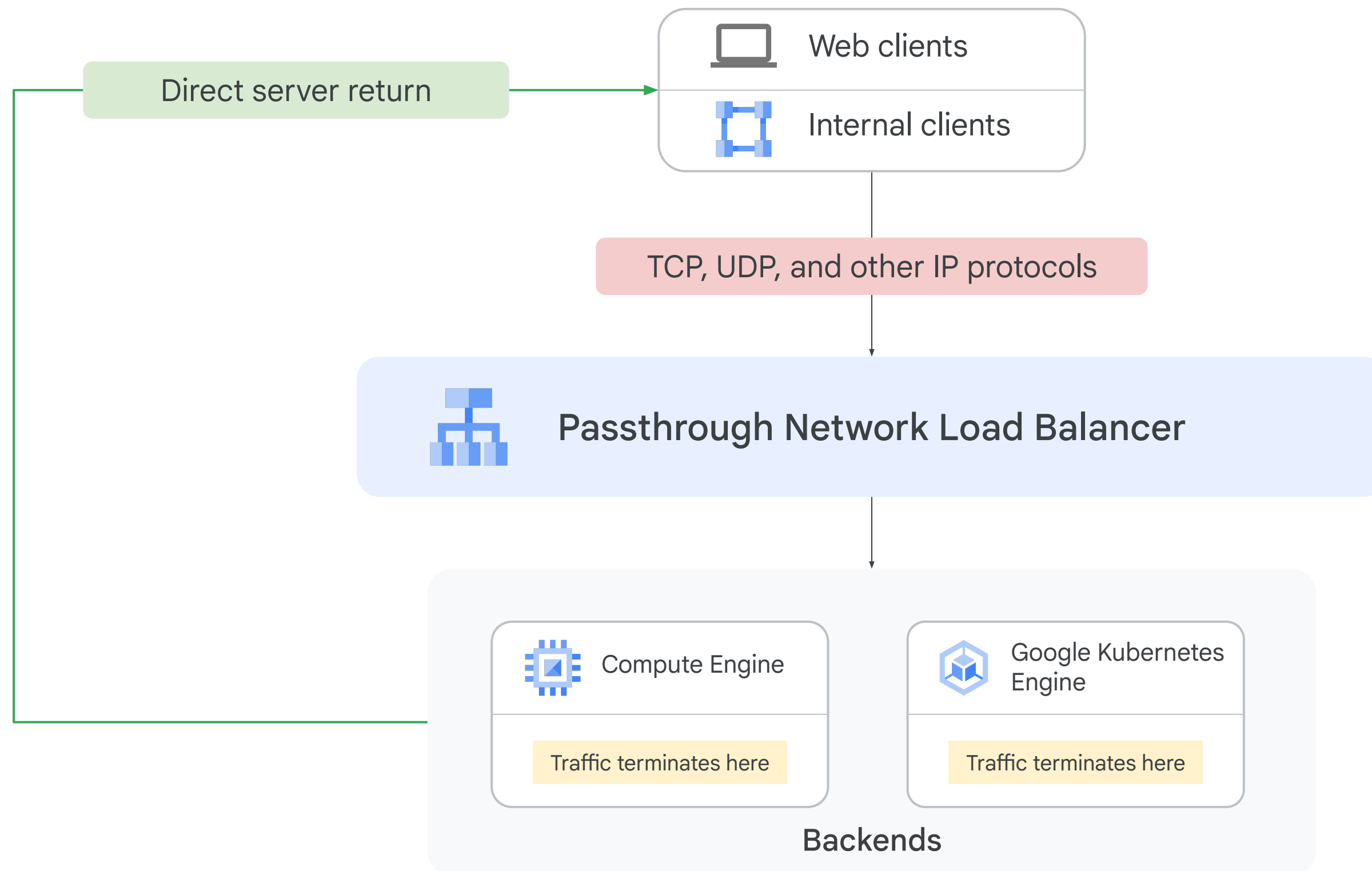
Proxy Network Load Balancer - Target TCP proxy



Proxy Network Load Balancer - Target SSL proxy



Architecture of a passthrough Network Load Balancer



Backend service-based architecture

- Regional backend service
- Defines the behavior of the load balancer and how it distributes traffic to its backend instance groups
- Support for IPv4 and IPv6 traffic
- Multiple protocols
- Managed and unmanaged instance groups
- Non legacy health checks

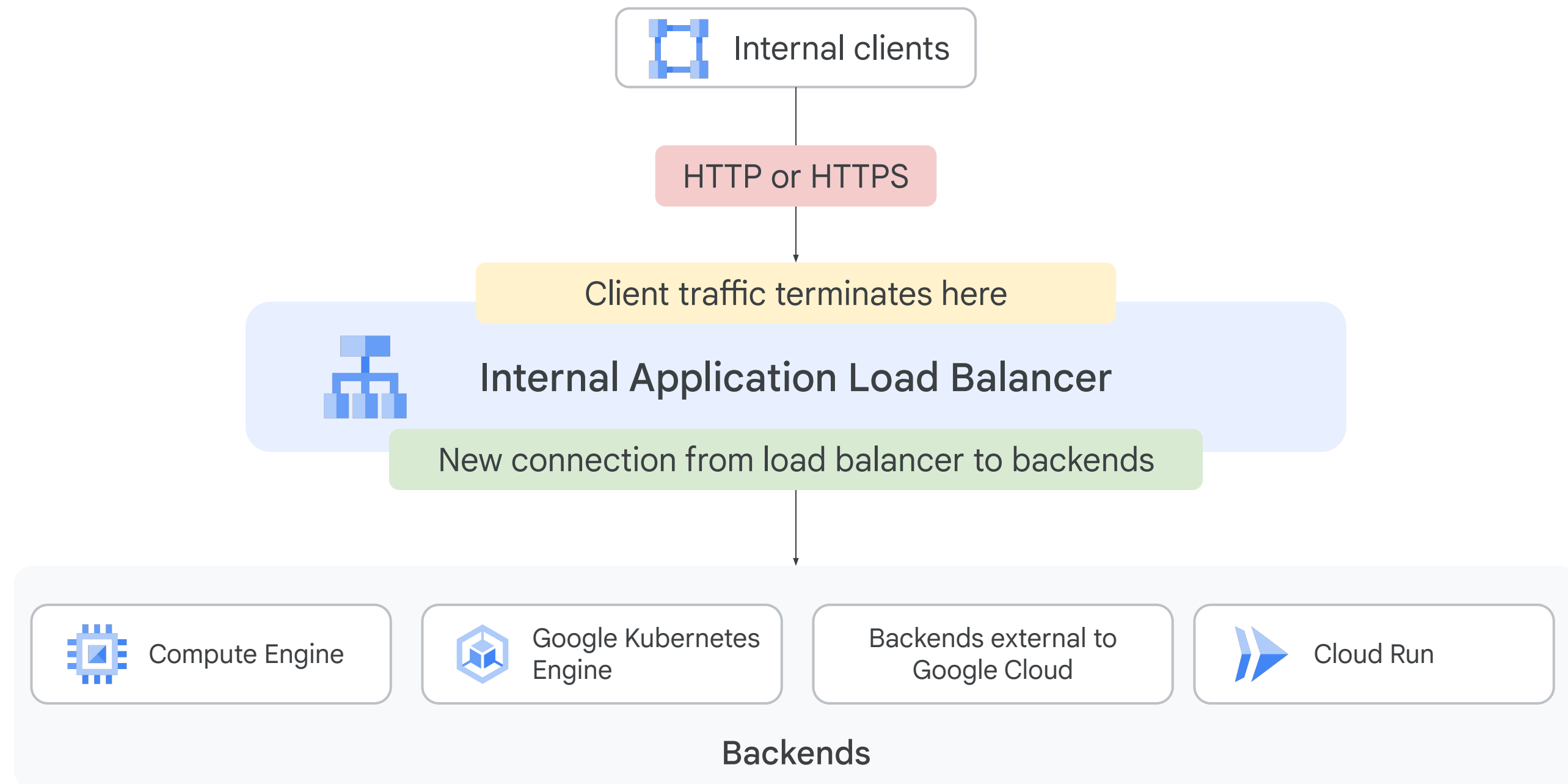
Target pool-based architecture

- Forwarding rules (TCP and UDP)
- Up to 50 per project
- One health check
- Instances must be in the same region



Internal Load Balancing

Architecture of an internal Application Load Balancer



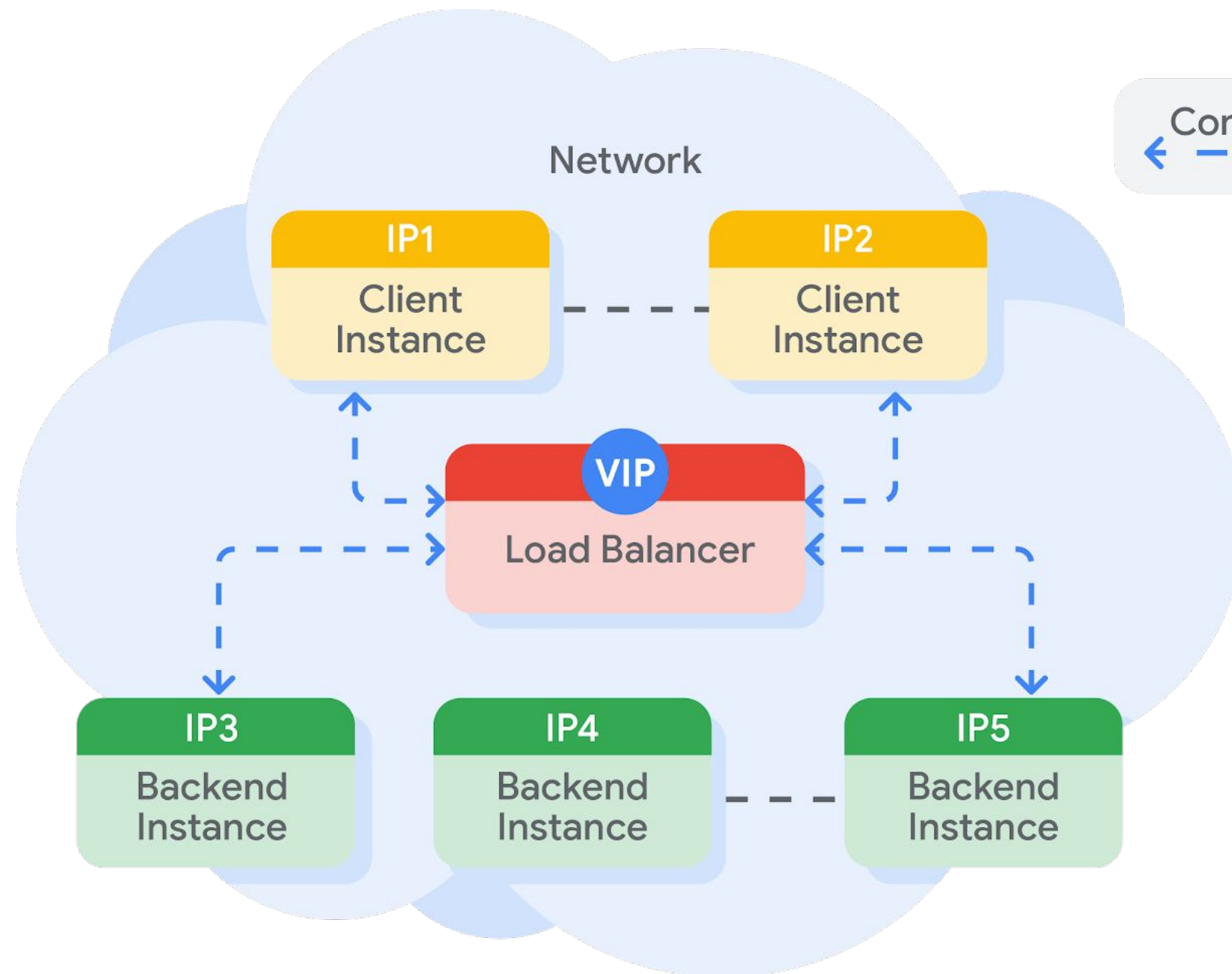
Internal Application Load Balancers

Deployment mode	Network service tier	Load balancing scheme	IP address	Frontend ports
Regional internal	Premium Tier	INTERNAL_MANAGED	IPv4	Can reference exactly one port from 1-65535
Cross-region internal	Premium Tier	INTERNAL_MANAGED	IPv4	

Internal passthrough Network Load Balancers

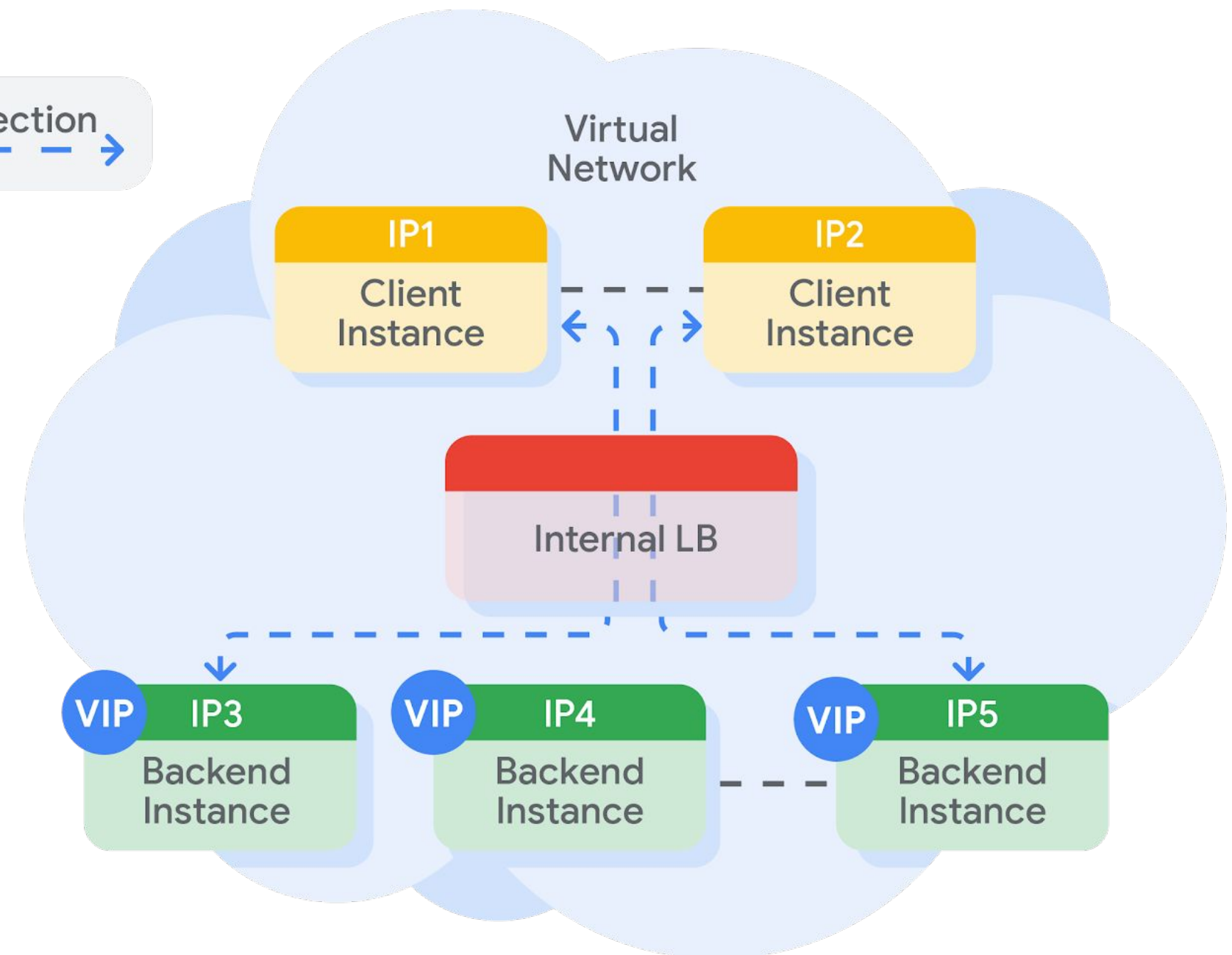
- Regional, private load balancing
 - VM instances in same region
 - RFC 1918 IP addresses
- TCP, UDP, ICMP, ICMPv6, SCTP, ESP, AH, and GRE traffic
- Reduced latency, simpler configuration
- Software-defined, fully distributed load balancing

Software-defined, fully distributed load balancing



Traditional proxy model of internal load balancing

Connection
← — — — →

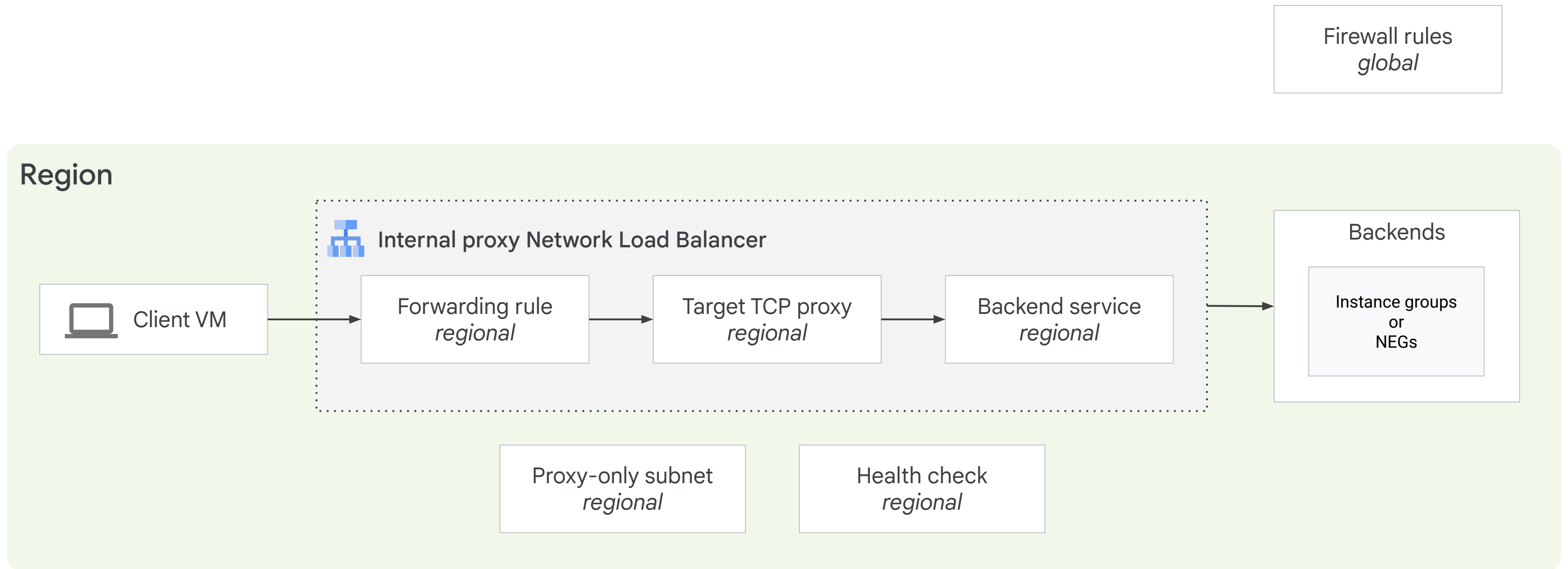


Internal passthrough Network Load Balancer

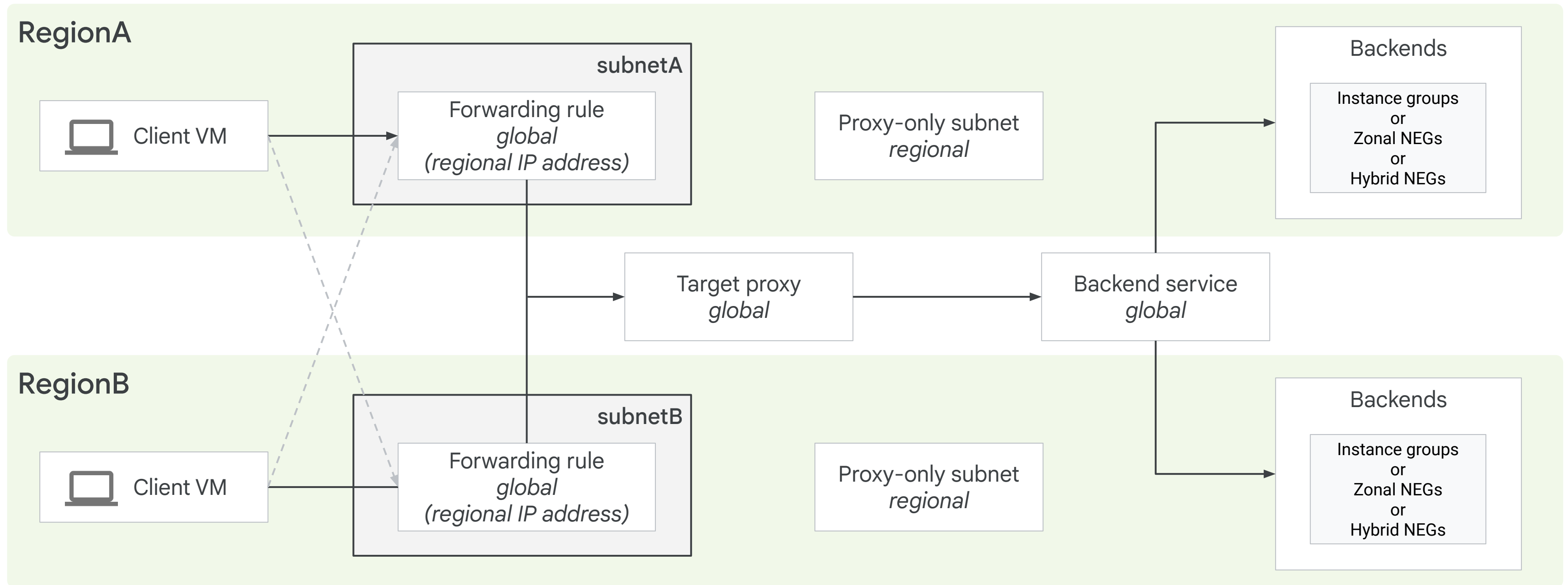
Internal proxy Network Load Balancers

- Proxy-based load balancer
- Balances traffic within your VPN network
- Regional or Cross-region
- Software-defined, fully distributed load balancing

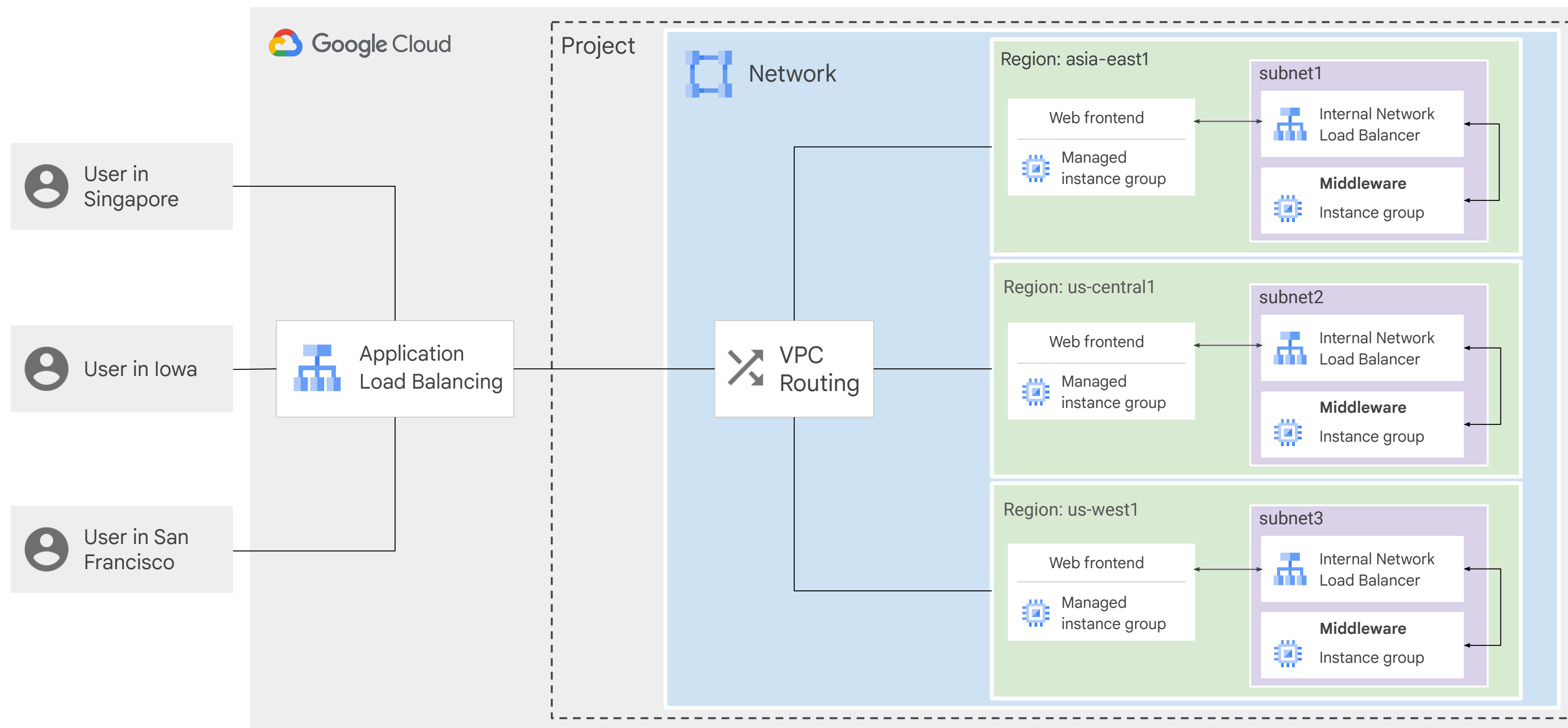
Architecture of a regional internal proxy Network Load Balancer



Architecture of a cross-region internal proxy Network Load Balancer

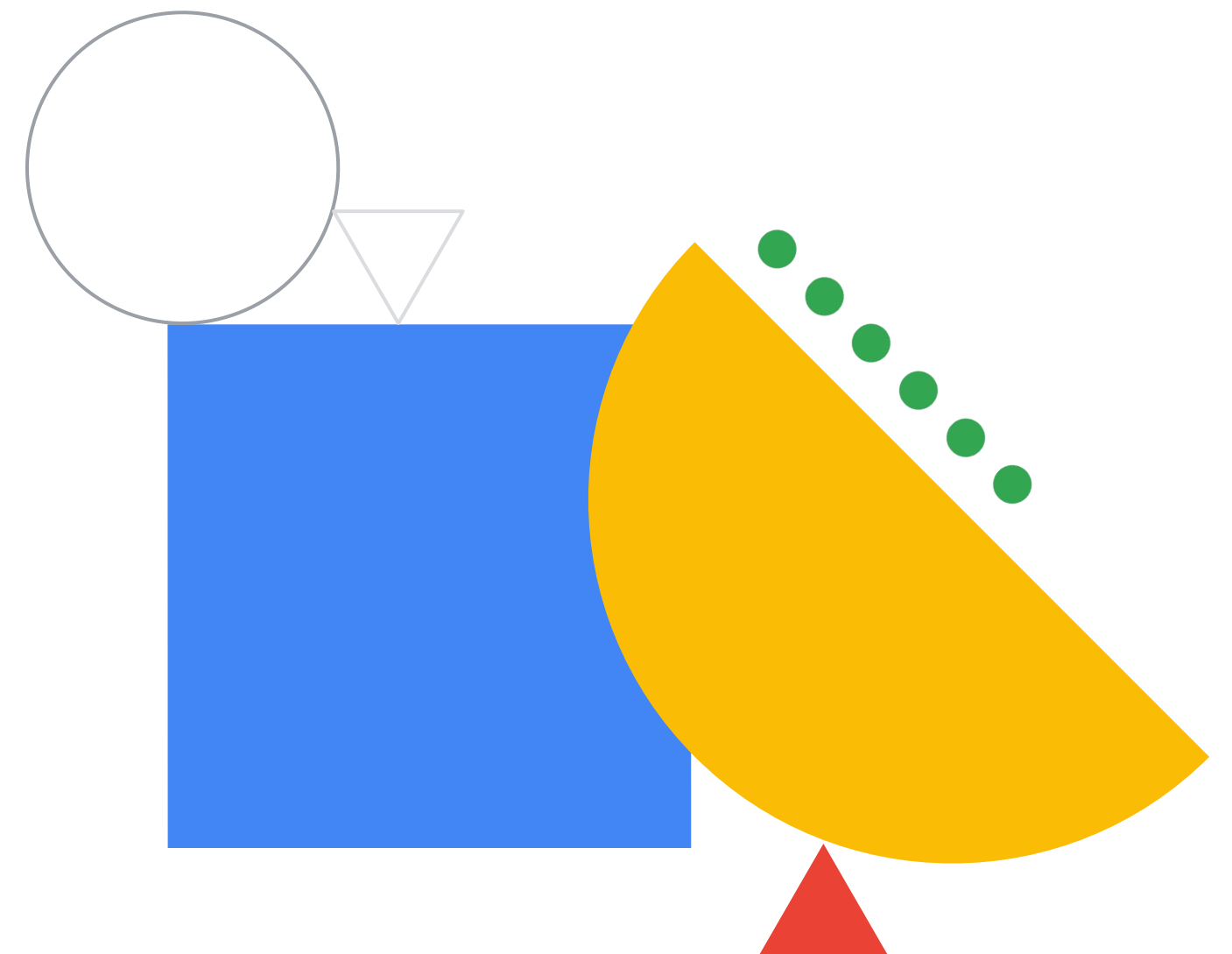


Internal load balancing supports 3-tier web services



Lab Intro

Configure an Internal Network
Load Balancer



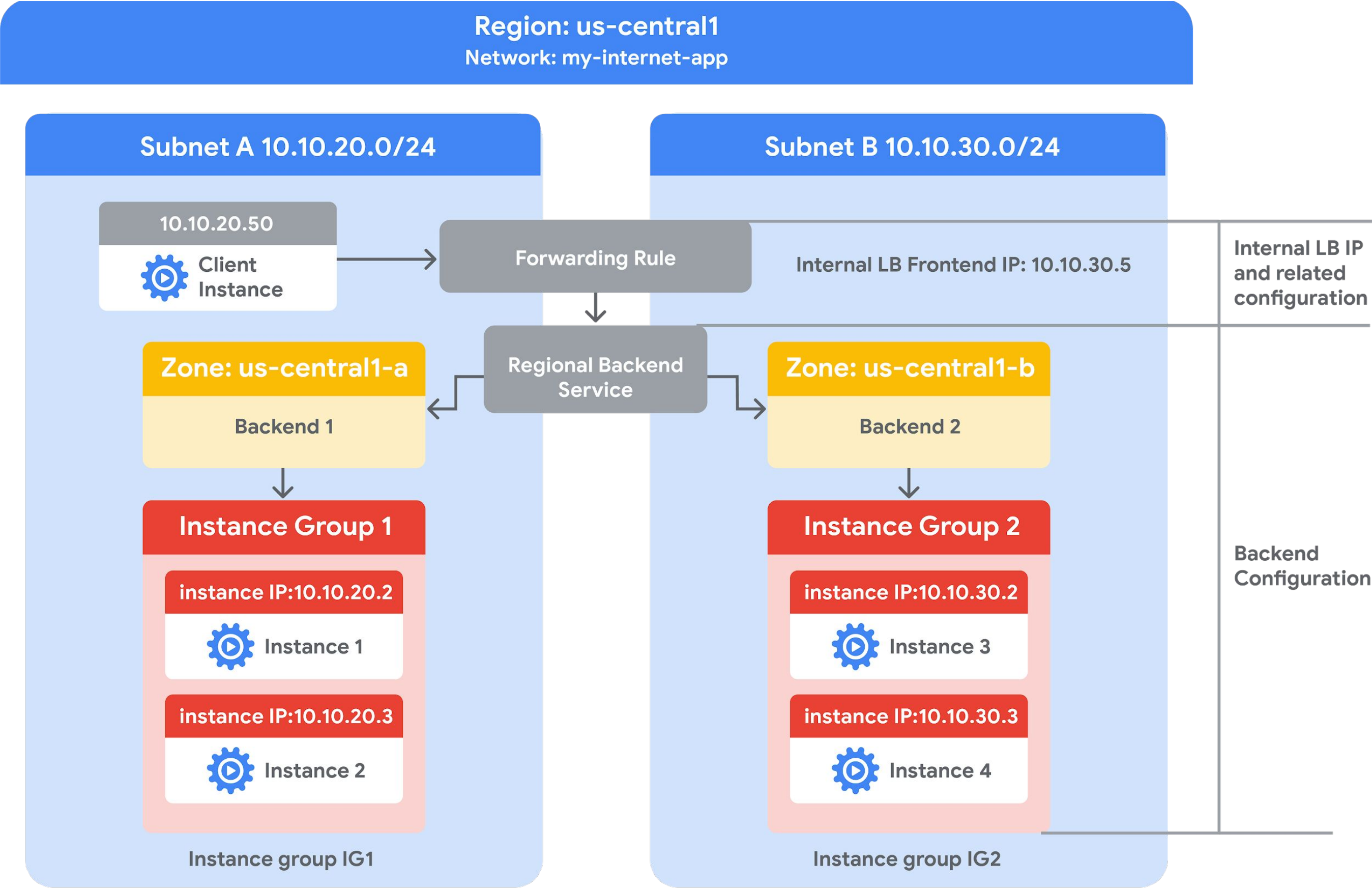
Lab objectives

01 Create HTTP and health check firewall rules

02 Configure two instance templates

03 Create two managed instance groups

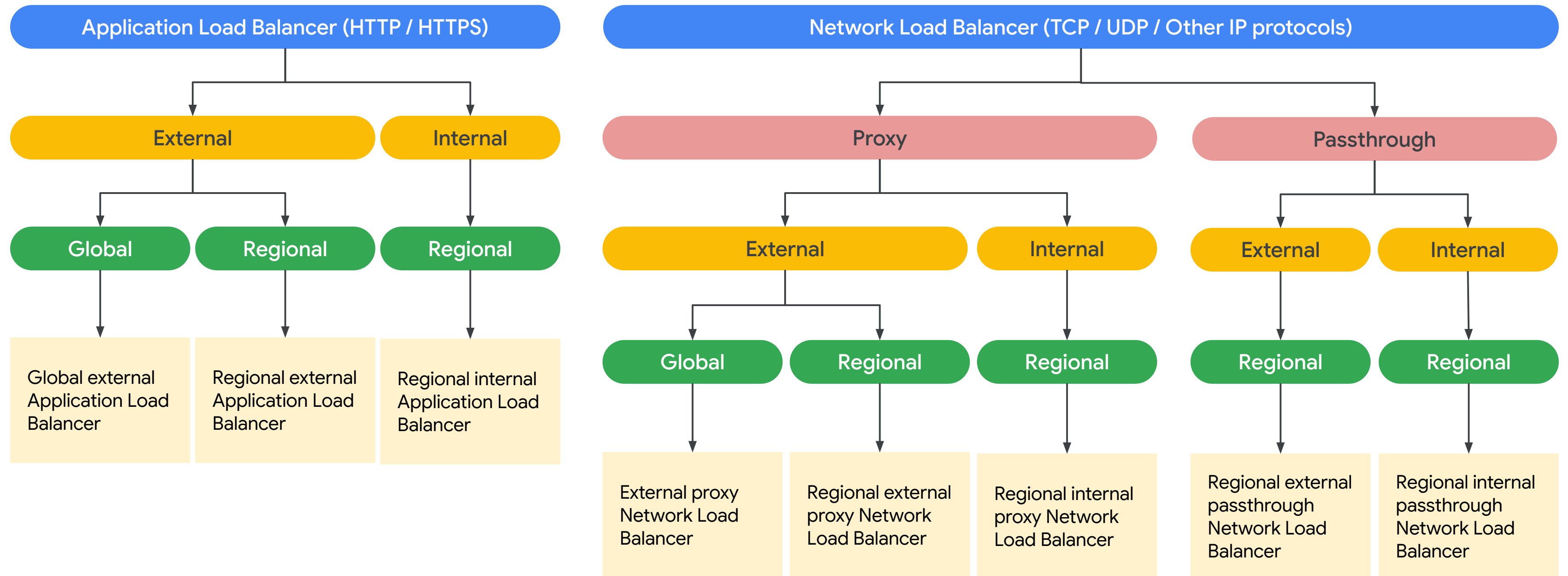






Choosing a Load Balancer

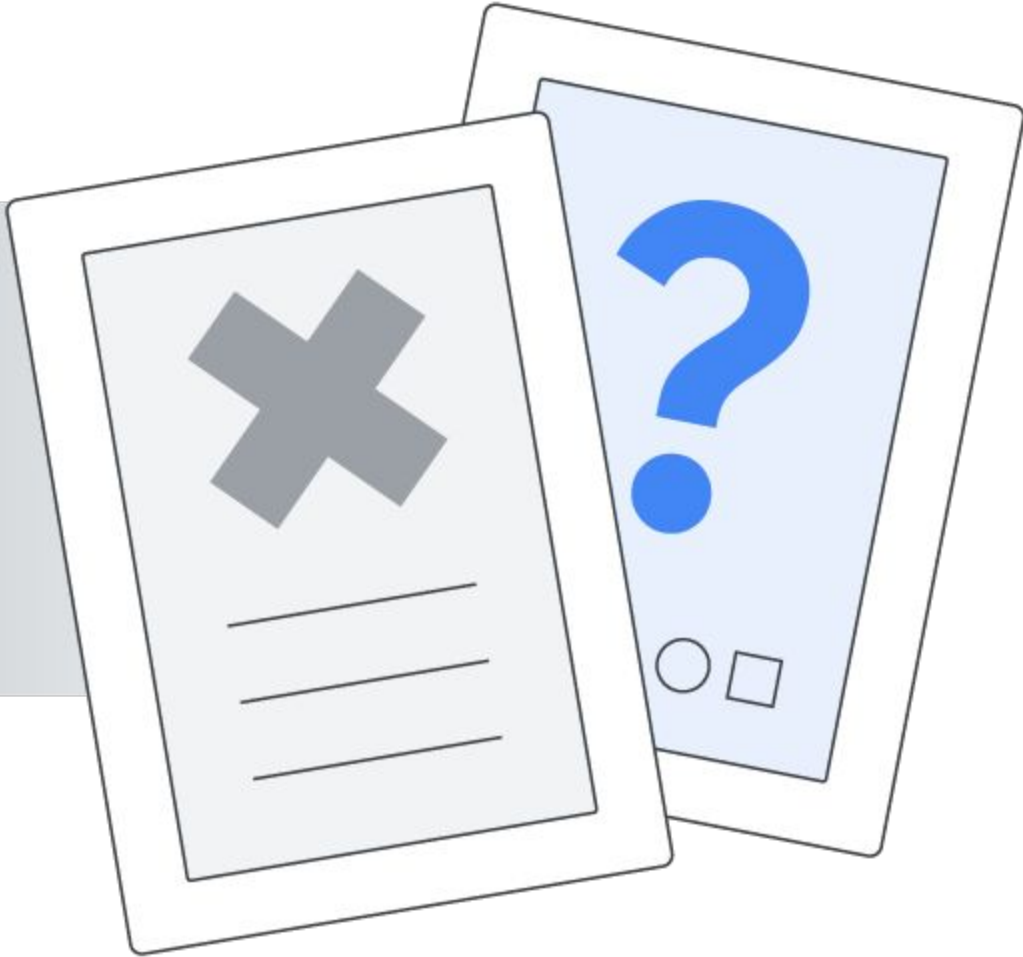
Deployment modes available for Cloud Load Balancing



Summary of Google Cloud load balancers

Load balancer	Deployment mode	Traffic type	Network Service Tier	Load-balancing scheme
Application Load Balancers	Global external	HTTP or HTTPS	Premium	EXTERNAL_MANAGED
	Regional external	HTTP or HTTPS	Standard	EXTERNAL_MANAGED
	Classic	HTTP or HTTPS	Global in Premium Regional in Standard	EXTERNAL
	Internal Always regional	HTTP or HTTPS	Premium	INTERNAL_MANAGED
Proxy Network Load Balancers	Global external	TCP with optional SSL offload	Global in Premium Regional in Standard	EXTERNAL
	Regional external	TCP	Standard only	EXTERNAL_MANAGED
	Internal Always regional	TCP without SSL offload	Premium only	INTERNAL_MANAGED
Passthrough Network Load Balancers	External Always regional	TCP, UDP, ESP, GRE, ICMP, and ICMPv6	Premium or Standard	EXTERNAL
	Internal Always regional	TCP or UDP	Premium only	INTERNAL

Quiz



Question #1

Question

Which of the following is not a Google Cloud load balancing service?

- A. Global external Application Load Balancer
- B. External proxy Network Load Balancer
- C. Regional external proxy Network Load Balancer
- D. Global hardware-defined Load Balancer
- E. Regional external passthrough Network Load Balancer
- F. Regional internal Application Load Balancer

Question #1

Answer

Which of the following is not a Google Cloud load balancing service?

- A. Global external Application Load Balancer
- B. External proxy Network Load Balancer
- C. Regional external proxy Network Load Balancer
- D. Global hardware-defined Load Balancer
- E. Regional external passthrough Network Load Balancer
- F. Regional internal Application Load Balancer



Question #2

Question

Which load balancer is recommended for HTTP(S) traffic?

- A. Regional internal passthrough Network Load Balancer
- B. Regional internal proxy Network Load Balancer
- C. Global external Application Load Balancer
- D. Global external proxy Network Load Balancer
- E. Regional external passthrough Network Load Balancer

Question #2

Answer

Which load balancer is recommended for HTTP(S) traffic?

- A. Regional internal passthrough Network Load Balancer
- B. Regional internal proxy Network Load Balancer
- C. Global external Application Load Balancer
- D. Global external proxy Network Load Balancer
- E. Regional external passthrough Network Load Balancer



Question #3

Question

Which of the following are applicable autoscaling policies for managed instance groups?

- A. CPU utilization
- B. Load balancing capacity
- C. Monitoring metrics
- D. Queue-based workload

Question #3

Answer

Which of the following are applicable autoscaling policies for managed instance groups?

- A. CPU utilization
- B. Load balancing capacity
- C. Monitoring metrics
- D. Queue-based workload



Review: Load Balancing and Autoscaling

